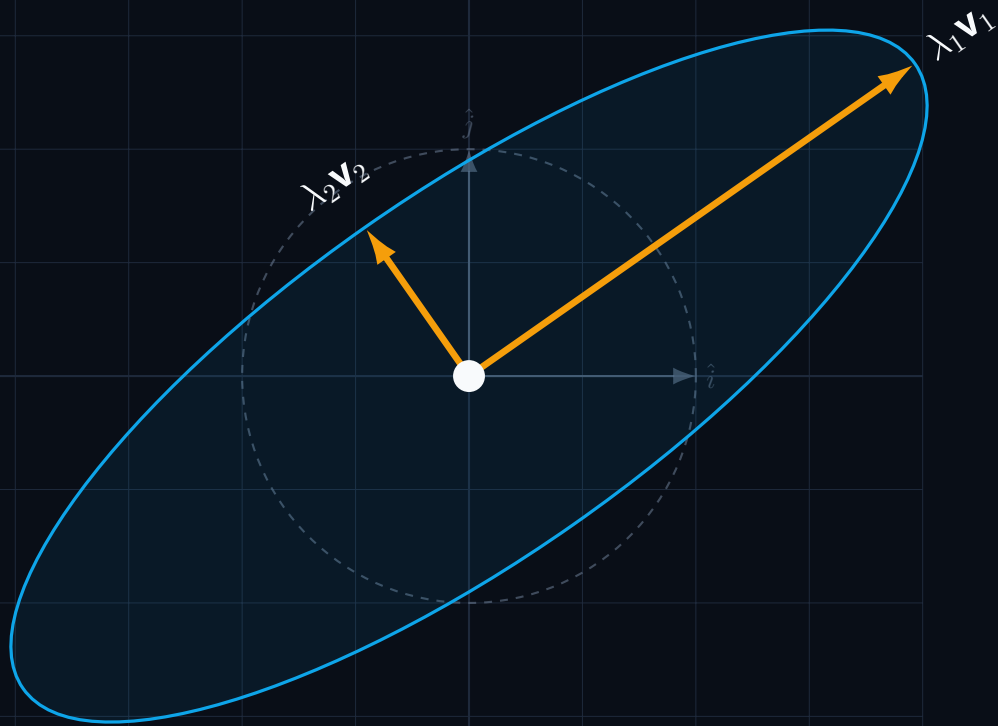


# LINEAR ALGEBRA



PREPARED & CURATED BY

**Shaikh Mahad**

GITHUB // [codewithmahad](#) LINKEDIN // [codewithmahad](#)

HUB

## Contents

|                                                                  |           |
|------------------------------------------------------------------|-----------|
| <b>Chapter 1: Foundational Algebraic Structures &amp; Axioms</b> | <b>2</b>  |
| Core Theory and Mechanisms . . . . .                             | 2         |
| Verifying Vector Space Axioms (Problem Bank) . . . . .           | 5         |
| <b>Chapter 2: Matrices and Determinants</b>                      | <b>9</b>  |
| Core Theory and Mechanisms . . . . .                             | 9         |
| Problem Bank (Assignment Solutions) . . . . .                    | 13        |
| Lecture Questions: Properties of Determinants . . . . .          | 18        |
| Lecture Questions: Matrix Operations & Applications . . . . .    | 21        |
| <b>Chapter 4: General Vector Spaces</b>                          | <b>26</b> |
| Key Concepts to Know, Revise, and Understand . . . . .           | 26        |
| Detailed Solutions to Exercises 4.2 & 4.3 . . . . .              | 28        |
| <b>Chapter 5: Eigenvalues and Eigenvectors</b>                   | <b>42</b> |
| Key Concepts to Know, Revise, and Understand . . . . .           | 42        |
| Detailed Solutions to Exercise 5.1 . . . . .                     | 44        |
| <b>About This Compilation</b>                                    | <b>54</b> |

## Chapter 1: Foundational Algebraic Structures & Vector Space Axioms

### Core Theory and Mechanisms

Before defining a Vector Space, we must understand the fundamental algebraic structures that build up to it: Groups, Rings, and Fields.

#### 1. Group & Abelian Group

**Group ( $G$ ):** A set combined with a binary operation (usually denoted by  $*$  or  $+$ ,  $\times$ ) that satisfies four fundamental axioms:

1. **Closure:** For all  $a, b \in G$ , the result of the operation  $a * b$  is also in  $G$ .
2. **Associativity:** For all  $a, b, c \in G$ ,  $(a * b) * c = a * (b * c)$ .
3. **Identity Element:** There exists an element  $e \in G$  such that for every  $a \in G$ ,  $e * a = a * e = a$ . (e.g., 0 for addition, 1 for multiplication).
4. **Inverse Element:** For each  $a \in G$ , there exists an inverse  $a^{-1} \in G$  such that  $a * a^{-1} = a^{-1} * a = e$ .

**Abelian Group (Commutative Group):** A group that satisfies one additional axiom:

5. **Commutativity:** For all  $a, b \in G$ ,  $a * b = b * a$ .

*Note: The set of vectors under vector addition forms an Abelian Group.*

## 2. Ring & Field

**Ring ( $R$ ):** A set equipped with *two* binary operations: Addition (+) and Multiplication ( $\times$ ). It must satisfy:

- $(R, +)$  is an **Abelian Group** under addition.
- $(R, \times)$  is associative under multiplication (Closure and Associativity).
- **Distributive Laws:** Multiplication distributes over addition. For all  $a, b, c \in R$ :

$$a \times (b + c) = (a \times b) + (a \times c) \quad \text{and} \quad (a + b) \times c = (a \times c) + (b \times c)$$

**Field ( $\mathbb{F}$ ):** A commutative ring with identity where *every non-zero element has a multiplicative inverse*. It essentially allows for the operations of addition, subtraction, multiplication, and division (except by zero).

- Common examples of fields are Real Numbers ( $\mathbb{R}$ ), Complex Numbers ( $\mathbb{C}$ ), and Rational Numbers ( $\mathbb{Q}$ ).

### Pro-Tip / Core Intuition

#### The Hierarchy of Algebraic Structures

Think of these mathematical structures like leveling up in a game:

**Group** (Basic: just +, -)  $\longrightarrow$  **Ring** (Adds  $\times$ )  $\longrightarrow$  **Field** (Adds  $\div$ ).

A Vector Space sits on top of this hierarchy. It needs a **Field** (like  $\mathbb{R}$ ) to provide its "scalars" (the numbers that scale the vectors).

### 3. Vector Space (Definition & 10 Axioms)

A **Vector Space**  $V$  over a field  $\mathbb{F}$  (usually  $\mathbb{R}$ ) is a set of elements called *vectors*, equipped with two operations: **Vector Addition** and **Scalar Multiplication**. To be a valid vector space, the set must satisfy the following 10 axioms for any vectors  $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$  and any scalars  $k, m \in \mathbb{F}$ :

#### Addition Axioms (Making $V$ an Abelian Group under $+$ ):

1. **Closure under Addition:**  $\mathbf{u} + \mathbf{v} \in V$
2. **Commutativity of Addition:**  $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3. **Associativity of Addition:**  $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
4. **Existence of Zero Vector:** There exists a  $\mathbf{0} \in V$  such that  $\mathbf{u} + \mathbf{0} = \mathbf{u}$ .
5. **Existence of Additive Inverse:** For every  $\mathbf{u} \in V$ , there is a  $-\mathbf{u} \in V$  such that  $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$ .

#### Scalar Multiplication Axioms:

6. **Closure under Scalar Multiplication:**  $k\mathbf{u} \in V$
7. **Distributivity of Scalar over Vector Addition:**  $k(\mathbf{u} + \mathbf{v}) = k\mathbf{u} + k\mathbf{v}$
8. **Distributivity of Vector over Scalar Addition:**  $(k + m)\mathbf{u} = k\mathbf{u} + m\mathbf{u}$
9. **Associativity of Scalar Multiplication:**  $k(m\mathbf{u}) = (km)\mathbf{u}$
10. **Scalar Identity:**  $1\mathbf{u} = \mathbf{u}$  (where 1 is the multiplicative identity of the field).

### Why This Matters: Real-World Applications

Where do Fields and Vector Spaces show up in modern tech?

- **Cryptography:** Modern encryption standards like AES rely heavily on *Galois Fields* (finite fields) because they allow mathematical operations (like shifting bits) that wrap around perfectly without losing information or causing overflow.
- **Quantum Computing:** Qubits exist in a highly complex vector space known as a Hilbert Space, where the "scalars" used to multiply the vectors come from the field of Complex Numbers ( $\mathbb{C}$ ).

## Verifying Vector Space Axioms (Problem Bank)

### Question 1

**Problem:** Let  $V$  be the set of ordered pairs of real numbers. Check whether  $V$  is a vector space or not over  $\mathbb{R}$ . If not, indicate the axiom which fails. The given operations are:

$$\begin{aligned}(a, b) + (c, d) &= (a + c, b + d) \\ k(a, b) &= (ka, 0)\end{aligned}$$

**Solution:**

#### 1. Checking Abelian group under (+):

**i) Closure Property:** If  $(a, b)$  and  $(c, d) \in V$ , then  $(a + c, b + d)$  also  $\in V$ . Closure property is satisfied!

**ii) Associative Property:** Let  $(a, b), (c, d)$  and  $(e, f) \in V$ .

$$\begin{aligned}[(a, b) + (c, d)] + (e, f) &= (a, b) + [(c, d) + (e, f)] \\ (a + c, b + d) + (e, f) &= (a, b) + (c + e, d + f) \\ (a + c + e, b + d + f) &= (a + c + e, b + d + f)\end{aligned}$$

Associative property satisfied.

**iii) Existence of Identity:** Let  $(0, 0) \in V$ , then:

$$(0, 0) + (a, b) = (0 + a, 0 + b) = (a, b)$$

The zero vector  $(0, 0)$  serves as the additive identity.

**iv) Existence of Inverse:** For every  $(a, b) \in V$ , there exists an element  $(-a, -b) \in V$  such that:

$$(a, b) + (-a, -b) = (a - a, b - b) = (0, 0)$$

$(-a, -b)$  is the additive inverse.

**v) Commutative Property:** Let  $(a, b), (c, d) \in V$ .

$$\begin{aligned}(a, b) + (c, d) &= (c, d) + (a, b) \\ (a + c, b + d) &= (a + c, b + d)\end{aligned}$$

Commutative Property satisfied.

**2. Distributive Law of Scalar over Vector Addition:** Let  $k \in \mathbb{R}$  be a scalar, and  $(a, b), (c, d) \in V$  be vectors.

$$\begin{aligned}\text{L.H.S} &= k[(a, b) + (c, d)] \\ &= k(a + c, b + d) \\ &= (k(a + c), 0) \\ &= (ka + kc, 0)\end{aligned}$$

$$\begin{aligned}\text{R.H.S} &= k(a, b) + k(c, d) \\ &= (ka, 0) + (kc, 0) \\ &= (ka + kc, 0 + 0) \\ &= (ka + kc, 0)\end{aligned}$$

L.H.S = R.H.S (Proved)

**3. Distributive Law of Vector over Scalar Addition:** Let  $k, l \in \mathbb{R}$  be scalars, and let  $(a, b) \in V$  be a vector.

$$\begin{aligned}\text{L.H.S} &= (k + l)(a, b) \\ &= ((k + l)a, 0) \\ &= (ka + la, 0)\end{aligned}$$

$$\begin{aligned}\text{R.H.S} &= k(a, b) + l(a, b) \\ &= (ka, 0) + (la, 0) \\ &= (ka + la, 0 + 0) \\ &= (ka + la, 0)\end{aligned}$$

L.H.S = R.H.S (Proved)

**4. Associativity of Scalar Multiplication:** Let  $k, l \in \mathbb{R}$  be scalars, and  $(a, b) \in V$  be a vector.

$$\begin{aligned}\text{L.H.S} &= k[l(a, b)] \\ &= k(la, 0) \\ &= (kla, 0)\end{aligned}$$

$$\begin{aligned}\text{R.H.S} &= (kl)(a, b) \\ &= (kla, 0)\end{aligned}$$

L.H.S = R.H.S (Proved)

**5. Scalar Identity:**

$$\begin{aligned}\text{L.H.S} &= 1(a, b) \\ &= (1a, 0) \\ &= (a, 0)\end{aligned}$$

$$\text{R.H.S} = (a, b)$$

L.H.S  $\neq$  R.H.S

**Conclusion:** The given set of elements of ordered pairs is **not** a vector space w.r.t the given operations because the Scalar Identity axiom fails.

## Question 2

**Problem:** Check whether  $V$  is a vector space or not over  $\mathbb{R}$  with the following given operations:

$$(a, b) + (c, d) = (a + c, b + d)$$

$$k(a, b) = (ka, kb)$$

**Solution:**

### 1. Checking Abelian group under (+):

**i) Closure Property:** Let  $(a, b)$  and  $(c, d) \in V$ , then  $(a + c, b + d)$  also  $\in V$ . Closure property is satisfied.

**ii) Associative Property:** Let  $(a, b), (c, d)$  and  $(e, f) \in V$ .

$$[(a, b) + (c, d)] + (e, f) = (a, b) + [(c, d) + (e, f)]$$

$$(a + c, b + d) + (e, f) = (a, b) + (c + e, d + f)$$

$$(a + c + e, b + d + f) = (a + c + e, b + d + f)$$

Associative property satisfied.

**iii) Existence of Identity:** Let  $(0, 0) \in V$ , then:

$$(0, 0) + (a, b) = (0 + a, 0 + b) = (a, b)$$

The zero vector  $(0, 0)$  serves as the additive identity.

**iv) Existence of Inverse:** For every  $(a, b) \in V$ , there exists an element  $(-a, -b) \in V$  such that:

$$(a, b) + (-a, -b) = (a - a, b - b) = (0, 0)$$

$(-a, -b)$  is the additive inverse.

**v) Commutative Property:** Let  $(a, b), (c, d) \in V$ .

$$(a, b) + (c, d) = (c, d) + (a, b)$$

$$(a + c, b + d) = (a + c, b + d)$$

Commutative Property satisfied.

**2. Distributive Law of Scalar over Vector Addition:** Let  $k \in \mathbb{R}$  be a scalar, and  $(a, b), (c, d) \in V$  be vectors.

$$\begin{aligned} \text{L.H.S} &= k[(a, b) + (c, d)] \\ &= k(a + c, b + d) \\ &= (k(a + c), k(b + d)) \\ &= (ka + kc, kb + kd) \end{aligned}$$



$$\begin{aligned}\text{R.H.S} &= k(a, b) + k(c, d) \\ &= (ka, kb) + (kc, kd) \\ &= (ka + kc, kb + kd)\end{aligned}$$

L.H.S = R.H.S (Proved)

**3. Distributive Law of Vector over Scalar Addition:** Let  $k, l \in \mathbb{R}$  be scalars, and let  $(a, b) \in V$  be a vector.

$$\begin{aligned}\text{L.H.S} &= (k + l)(a, b) \\ &= ((k + l)a, (k + l)b) \\ &= (ka + la, kb + lb)\end{aligned}$$

$$\begin{aligned}\text{R.H.S} &= k(a, b) + l(a, b) \\ &= (ka, kb) + (la, lb) \\ &= (ka + la, kb + lb)\end{aligned}$$

L.H.S = R.H.S (Proved)

**4. Associativity of Scalar Multiplication:** Let  $k, l \in \mathbb{R}$  be scalars, and  $(a, b) \in V$  be a vector.

$$\begin{aligned}\text{L.H.S} &= k[l(a, b)] \\ &= k(la, lb) \\ &= (kla, klb)\end{aligned}$$

$$\begin{aligned}\text{R.H.S} &= (kl)(a, b) \\ &= (kla, klb)\end{aligned}$$

L.H.S = R.H.S (Proved)

**5. Scalar Identity:**

$$\begin{aligned}\text{L.H.S} &= 1(a, b) \\ &= (1a, 1b) \\ &= (a, b)\end{aligned}$$

$$\text{R.H.S} = (a, b)$$

L.H.S = R.H.S (Proved)

**Conclusion:** Since the set satisfies all 10 axioms,  $V$  is a valid vector space over  $\mathbb{R}$ .

## Chapter 2: Matrices and Determinants

### Core Theory and Mechanisms

*A comprehensive, step-by-step revision guide covering fundamental properties, operations, and theorem applications.*

#### Fundamental Types of Matrices (In Plain English)

- **Square Matrix** ( $n \times n$ ): Equal rows and columns. *(Only square matrices can have determinants and inverses!)*
- **Diagonal Matrix**: All entries outside the main diagonal are strictly zero. *(Super fast to multiply and find powers of!)*
- **Identity Matrix** ( $I$ ): The matrix equivalent of the number "1". Main diagonal is all 1s. ( $AI = IA = A$ ).
- **Symmetric Matrix** ( $A = A^T$ ): The matrix mirrors itself perfectly across the main diagonal.
- **Skew-Symmetric Matrix** ( $A = -A^T$ ): Mirrors itself with flipped signs. Its main diagonal *must* be all zeros.
- **Singular Matrix** ( $\det(A) = 0$ ): A "destructive" matrix. It collapses space, destroying information, which means it **cannot be inverted**.
- **Non-Singular Matrix** ( $\det(A) \neq 0$ ): Space remains intact (though stretched/rotated), meaning it **is invertible**.

### The Core Duality: Matrix vs. Determinant

It is crucial to distinguish between matrix operations (applying to the entire array) and determinant properties (applying to scalar values).

| Operation                     | Matrix (The Array)                     | Determinant (The Scalar)                |
|-------------------------------|----------------------------------------|-----------------------------------------|
| Scalar Multiplication ( $k$ ) | Multiplies <b>every</b> element.       | Multiplies <b>only one</b> row/col.     |
| Taking a Common Factor        | Factored from <b>every</b> element.    | Factored from <b>a single</b> row/col.  |
| Multiplying a Row by $-1$     | Permanently changes matrix values.     | Must extract $-1$ outside: $-\det(A)$ . |
| Swapping Rows/-Columns        | Creates a completely different matrix. | Multiplies the determinant by $-1$ .    |

### □ Class Topper Example: Factoring

- **In a Matrix:** You pull the common factor out from *every single element*.

$$\begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix} = 2 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

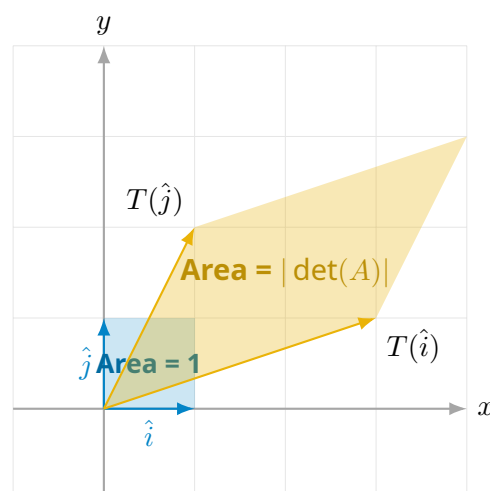
- **In a Determinant:** You pull the common factor out from *just one row (or column)* at a time!

$$\begin{vmatrix} 2 & 4 \\ 6 & 8 \end{vmatrix} = 2 \times \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} \text{ (pulled 2 from Row 1)} = 2 \times 2 \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 4 \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$$

### Pro-Tip / Core Intuition

#### Geometric Intuition of the Determinant

Algebraically, a determinant is a number computed from a square matrix. **Geometrically, it represents the scaling factor of a linear transformation.** If you apply a  $2 \times 2$  matrix transformation to the space, the determinant tells you how much the area of a unit square increases or decreases. If  $\det(A) = 0$ , the space has collapsed into a lower dimension (e.g., a 2D plane squashed into a 1D line), which is why the matrix is non-invertible!



## Matrix Arithmetic and Inverses

Matrices follow strict dimension constraints. **Matrix multiplication is NOT commutative** ( $AB \neq BA$ ).

### Solving for an Unknown Matrix:

Matrix division does not exist. You must multiply by the **Inverse** ( $A^{-1}$ ), respecting left-multiplication vs. right-multiplication.

- **Left-Multiply:** If  $AB = C$ , multiply both sides on the *left* by  $A^{-1}$ :

$$A^{-1}(AB) = A^{-1}C \implies IB = A^{-1}C \implies \mathbf{B} = \mathbf{A}^{-1}\mathbf{C}$$

- **Right-Multiply:** If  $XA = C$ , multiply both sides on the *right* by  $A^{-1}$ :

$$(XA)A^{-1} = C(A^{-1}) \implies XI = CA^{-1} \implies \mathbf{X} = \mathbf{C}\mathbf{A}^{-1}$$

## Row Echelon Form (REF) and Matrix Rank

**Rank:** The number of linearly independent rows in a matrix.

### Row Echelon Form (REF) & RREF

#### Row Echelon Form (REF):

To reach strict Row Echelon Form, use elementary row operations to create zeros below the leading entry of each row, **and scale the rows so every leading entry (pivot) is exactly 1**. Stop once a staircase form is achieved to find the rank or solve via back-substitution.

#### Reduced Row Echelon Form (RREF):

Building on REF, further operations are applied to zero out entries *above* the leading 1s. This is used for finding inverses via Gauss-Jordan elimination.

## Determinant Evaluation Techniques

**1. Arrow Technique (Sarrus' Rule):** Valid *only* for  $3 \times 3$  matrices. Recopy the first two columns to the right. Sum the products of the three downward diagonals, and subtract the products of the three upward diagonals.

**2. Cofactor Expansion:** Expand along any row or column. Multiply each element by its cofactor  $C_{ij} = (-1)^{i+j}M_{ij}$ . The signs follow a checkerboard pattern (starting with + in the top-left).

**3. Triangular Matrix Method:** Use row operations ( $R_i \rightarrow R_i - kR_j$ ) to create zeros below the main diagonal. The determinant is simply the **product of the main diagonal elements**.

## Advanced Determinant Properties & Identities

### Standard Properties:

1. **Zero Property:** If any row/column is all zeros,  $\det(A) = 0$ .
2. **Identical Property:** If two rows/columns are identical,  $\det(A) = 0$ .
3. **Invariance Property:**  $R_i \rightarrow R_i + kR_j$  **does not** change the determinant.

### Vandermonde Determinant:

A highly specific pattern where columns (or rows) follow increasing powers.

$$\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} = (x-y)(y-z)(z-x)$$

### Complex Roots of Unity:

If  $\omega$  is the  $n$ -th root of unity ( $\omega^n = 1$ ), the sum of its consecutive powers equals zero:  $1 + \omega + \omega^2 + \dots + \omega^{n-1} = 0$ . This is useful for collapsing large characteristic matrices.

## Calculating the Inverse Matrix

**Formula Method:**  $A^{-1} = \frac{1}{\det(A)} \text{Adj}(A)$

- **Adjoint** ( $3 \times 3$ ): Find the Matrix of Minors, apply the alternating Sign Matrix (+ - +), and **Transpose** the result.

### Gauss-Jordan Method:

Set up  $[A \mid I]$ . Perform row operations to reach Reduced Row Echelon Form (RREF) until the left side is  $I$ . The right side becomes  $A^{-1}$ .

## Problem Bank (Assignment Solutions)

*Selected and solved questions from Elementary Linear Algebra: Applications Version by Howard Anton & Chris Rorres.*

### Exercise 2.1 Solutions

**Q7. Evaluate the determinant of the given matrix. If the matrix is invertible, use Equation (2) to find its inverse.**

$$A = \begin{bmatrix} -5 & 7 \\ -7 & -2 \end{bmatrix}$$

**Solution:**

Evaluating the Determinant:

$$\det(A) = (-5)(-2) - (7)(-7) = 10 + 49 = 59$$

Since  $\det(A) \neq 0$ , the matrix is invertible.

Using the inverse formula for a  $2 \times 2$  matrix  $A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ :

$$A^{-1} = \frac{1}{59} \begin{bmatrix} -2 & -7 \\ 7 & -5 \end{bmatrix} = \begin{bmatrix} -2/59 & -7/59 \\ 7/59 & -5/59 \end{bmatrix}$$

**Q13. Use the arrow technique to evaluate the determinant.**

$$A = \begin{bmatrix} 3 & 0 & 0 \\ 2 & -1 & 5 \\ 1 & 9 & -4 \end{bmatrix}$$

**Solution:**

Recopying the first and second columns to the right:

$$\begin{array}{ccc|cc} 3 & 0 & 0 & 3 & 0 \\ 2 & -1 & 5 & 2 & -1 \\ 1 & 9 & -4 & 1 & 9 \end{array}$$

Summing the products of downward arrows and subtracting upward arrows:

$$\begin{aligned} \det(A) &= [(3)(-1)(-4) + (0)(5)(1) + (0)(2)(9)] - [(1)(-1)(0) + (9)(5)(3) + (-4)(2)(0)] \\ &= (12 + 0 + 0) - (0 + 135 + 0) \\ &= 12 - 135 \\ &= -123 \end{aligned}$$

**Q16. Find all values of  $\lambda$  for which  $\det(A) = 0$ .**

$$A = \begin{bmatrix} \lambda - 4 & 0 & 0 \\ 0 & \lambda & 2 \\ 0 & 3 & \lambda - 1 \end{bmatrix}$$

**Solution:**

Evaluating  $\det(A) = 0$  by cofactor expansion along the first row ( $R_1$ ):

$$\begin{aligned}(\lambda - 4) \begin{vmatrix} \lambda & 2 \\ 3 & \lambda - 1 \end{vmatrix} - 0 + 0 &= 0 \\(\lambda - 4)[\lambda(\lambda - 1) - (2)(3)] &= 0 \\(\lambda - 4)[\lambda^2 - \lambda - 6] &= 0\end{aligned}$$

Factoring the quadratic by middle-term break:

$$\begin{aligned}(\lambda - 4)[\lambda^2 - 3\lambda + 2\lambda - 6] &= 0 \\(\lambda - 4)[\lambda(\lambda - 3) + 2(\lambda - 3)] &= 0 \\(\lambda - 4)(\lambda + 2)(\lambda - 3) &= 0\end{aligned}$$

**Answer:**  $\lambda = 4, \lambda = -2, \lambda = 3$

**Q24. Evaluate  $\det(A)$  by a cofactor expansion along a row or column of your choice.**

$$A = \begin{bmatrix} k+1 & k-1 & 7 \\ 2 & k-3 & 4 \\ 5 & k+1 & k \end{bmatrix}$$

**Solution:**

Expanding along the second row ( $R_2$ ) applying the  $(- + -)$  checkerboard signs:

$$\begin{aligned}\det(A) &= -2 \begin{vmatrix} k-1 & 7 \\ k+1 & k \end{vmatrix} + (k-3) \begin{vmatrix} k+1 & 7 \\ 5 & k \end{vmatrix} - 4 \begin{vmatrix} k+1 & k-1 \\ 5 & k+1 \end{vmatrix} \\&= -2[k(k-1) - 7(k+1)] + (k-3)[k(k+1) - 35] - 4[(k+1)^2 - 5(k-1)] \\&= -2(k^2 - 8k - 7) + (k-3)(k^2 + k - 35) - 4(k^2 - 3k + 6) \\&= (-2k^2 + 16k + 14) + (k^3 - 2k^2 - 38k + 105) - (4k^2 - 12k + 24) \\&= \mathbf{k^3 - 8k^2 - 10k + 95}\end{aligned}$$

**Exercise 2.2 Solutions**

**Q12. Evaluate the determinant by first reducing the matrix to row echelon form and then using some combination of row operations and cofactor expansion.**

$$A = \begin{bmatrix} 1 & -3 & 0 \\ -2 & 4 & 1 \\ 5 & -2 & 2 \end{bmatrix}$$

**Solution:**

Applying row operations to introduce zeroes in the first column:

$$\det(A) = \begin{vmatrix} 1 & -3 & 0 \\ -2 & 4 & 1 \\ 5 & -2 & 2 \end{vmatrix} \begin{matrix} R_2 \rightarrow R_2 + 2R_1 \\ R_3 \rightarrow R_3 - 5R_1 \end{matrix} = \begin{vmatrix} 1 & -3 & 0 \\ 0 & -2 & 1 \\ 0 & 13 & 2 \end{vmatrix}$$

Evaluating by cofactor expansion along the first column:

$$= 1 \cdot \begin{vmatrix} -2 & 1 \\ 13 & 2 \end{vmatrix} - 0 + 0 = (-4) - (13) = -17$$

**Q14. Evaluate the determinant.**

$$A = \begin{bmatrix} 1 & -2 & 3 & 1 \\ 5 & -9 & 6 & 3 \\ -1 & 2 & -6 & -2 \\ 2 & 8 & 6 & 1 \end{bmatrix}$$

**Solution:**

Applying row operations to introduce zeroes in the first column:

$$\det(A) = \begin{vmatrix} 1 & -2 & 3 & 1 \\ 5 & -9 & 6 & 3 \\ -1 & 2 & -6 & -2 \\ 2 & 8 & 6 & 1 \end{vmatrix} \begin{matrix} R_2 \rightarrow R_2 - 5R_1 \\ R_3 \rightarrow R_3 + R_1 \\ R_4 \rightarrow R_4 - 2R_1 \end{matrix} = \begin{vmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & -9 & -2 \\ 0 & 0 & -3 & -1 \\ 0 & 12 & 0 & -1 \end{vmatrix}$$

Evaluating by cofactor expansion along  $C_1$ :

$$= 1 \cdot \begin{vmatrix} 1 & -9 & -2 \\ 0 & -3 & -1 \\ 12 & 0 & -1 \end{vmatrix} R_3 \rightarrow R_3 - 12R_1 = \begin{vmatrix} 1 & -9 & -2 \\ 0 & -3 & -1 \\ 0 & 108 & 23 \end{vmatrix}$$

Evaluating by cofactor expansion along the first column of the minor:

$$= 1 \cdot \begin{vmatrix} -3 & -1 \\ 108 & 23 \end{vmatrix} = (-3)(23) - (-1)(108) = -69 + 108 = 39$$

**Q18. Evaluate the determinant, given that:**  $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -6$

$$\text{Evaluate: } \begin{vmatrix} a+d & b+e & c+f \\ -d & -e & -f \\ g & h & i \end{vmatrix}$$

**Solution:**

Factoring out the scalar  $-1$  from row 2 ( $R_2$ ):

$$= -1 \begin{vmatrix} a+d & b+e & c+f \\ d & e & f \\ g & h & i \end{vmatrix} R_1 \rightarrow R_1 - R_2 = -1 \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

Substituting the known value of the determinant:

$$= -1(-6) = 6$$



**Q29. Show that  $\det(A) = 0$  without directly evaluating the determinant.**

$$A = \begin{bmatrix} -2 & 8 & 1 & 4 \\ 3 & 2 & 5 & 1 \\ 1 & 10 & 6 & 5 \\ 4 & -6 & 4 & -3 \end{bmatrix}$$

**Solution:**

Notice that adding the corresponding entries of  $R_1$  and  $R_2$  yields exactly  $R_3$ :

$$R_1 + R_2 = [1 \quad 10 \quad 6 \quad 5] = R_3$$

Applying row operations to zero out  $R_3$ :

$$\det(A) = \begin{vmatrix} -2 & 8 & 1 & 4 \\ 3 & 2 & 5 & 1 \\ 1 & 10 & 6 & 5 \\ 4 & -6 & 4 & -3 \end{vmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1 - R_2} \begin{vmatrix} -2 & 8 & 1 & 4 \\ 3 & 2 & 5 & 1 \\ 0 & 0 & 0 & 0 \\ 4 & -6 & 4 & -3 \end{vmatrix}$$

If a square matrix has a row of zeros, its determinant is exactly 0. Therefore,  $\det(A) = 0$ .

### Exercise 2.3 Solutions

**Q12. Use determinants to decide whether the given matrix is invertible.**

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 9 & -1 & 4 \\ 8 & 9 & -1 \end{bmatrix}$$

**Solution:**

Determinant test for invertibility. Evaluating by cofactor expansion along the first row:

$$\det(A) = 1 \begin{vmatrix} -1 & 4 \\ 9 & -1 \end{vmatrix} - 0 + (-1) \begin{vmatrix} 9 & -1 \\ 8 & 9 \end{vmatrix}$$

$$\det(A) = 1(1 - 36) - 1(81 + 8) = -35 - 89 = -124$$

A square matrix is invertible if and only if  $\det(A) \neq 0$ . Since  $\det(A) = -124$ , matrix  $A$  is invertible.

**Q18. Find the values of  $k$  for which the matrix  $A$  is invertible.**

$$A = \begin{bmatrix} 1 & 2 & 0 \\ k & 1 & k \\ 0 & 2 & 1 \end{bmatrix}$$

**Solution:**

Evaluating the determinant in terms of  $k$  along the first row:

$$\det(A) = 1 \begin{vmatrix} k & k \\ 0 & 1 \end{vmatrix} - 2 \begin{vmatrix} k & k \\ 0 & 1 \end{vmatrix} + 0 = (1 - 2k) - 2(k) = 1 - 4k$$

The matrix is invertible if and only if  $\det(A) \neq 0$ .

$$1 - 4k \neq 0 \implies 1 \neq 4k \implies k \neq 1/4$$

The matrix  $A$  is invertible for all real values of  $k$  except  $k = 1/4$ .

**Q20. Decide whether the matrix is invertible and if so, use the adjoint method to find its inverse.**

$$A = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 3 & 2 \\ -2 & 0 & -4 \end{bmatrix}$$

**Solution:**

Determinant Test:

$$\det(A) = 2(-12 - 0) - 0 + 3(0 - (-6)) = -24 + 18 = -6$$

Since  $\det(A) \neq 0$ , the matrix is invertible.

Calculating the Cofactor Matrix  $C_{ij} = (-1)^{i+j}M_{ij}$ :

$$\begin{array}{lll} C_{11} = +(-12 - 0) = -12 & C_{12} = -(0 - (-4)) = -4 & C_{13} = +(0 - (-6)) = 6 \\ C_{21} = -(0 - 0) = 0 & C_{22} = +(-8 - (-6)) = -2 & C_{23} = -(0 - 0) = 0 \\ C_{31} = +(0 - 9) = -9 & C_{32} = -(4 - 0) = -4 & C_{33} = +(6 - 0) = 6 \end{array}$$

$$\text{Matrix of Cofactors} = \begin{bmatrix} -12 & -4 & 6 \\ 0 & -2 & 0 \\ -9 & -4 & 6 \end{bmatrix}$$

$$\text{Adjoint (Transpose of Cofactor Matrix): } \text{adj}(A) = \begin{bmatrix} -12 & 0 & -9 \\ -4 & -2 & -4 \\ 6 & 0 & 6 \end{bmatrix}$$

Using  $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$ :

$$A^{-1} = \frac{1}{-6} \begin{bmatrix} -12 & 0 & -9 \\ -4 & -2 & -4 \\ 6 & 0 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 3/2 \\ 2/3 & 1/3 & 2/3 \\ -1 & 0 & -1 \end{bmatrix}$$

## Lecture Questions: Properties of Determinants

Advanced proofs and problem-solving strategies covered in lectures.

**Q1. Evaluate the value of  $x$  for which the determinant equals 0.**

$$\begin{vmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & x+1 & 3 & 4 & 5 \\ 1 & 2 & x+1 & 4 & 5 \\ 1 & 2 & 3 & x+1 & 5 \\ 1 & 2 & 3 & 4 & x+1 \end{vmatrix} = 0$$

**Solution:**

Applying row operations to zero out the first column:

$$\begin{matrix} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \\ R_4 \rightarrow R_4 - R_1 \\ R_5 \rightarrow R_5 - R_1 \end{matrix} \begin{vmatrix} 1 & 2 & 3 & 4 & 5 \\ 0 & x-1 & 0 & 0 & 0 \\ 0 & 0 & x-2 & 0 & 0 \\ 0 & 0 & 0 & x-3 & 0 \\ 0 & 0 & 0 & 0 & x-4 \end{vmatrix} = 0$$

Since the resulting matrix is lower triangular, the determinant is the product of the diagonal entries:

$$1 \cdot (x-1)(x-2)(x-3)(x-4) = 0$$

**Answer:  $x = 1, x = 2, x = 3, x = 4$ .**

**Q2. Prove that the given determinant is a multiple of the matrix below and find the multiplier.**

$$\begin{vmatrix} 2b_1 + c_1 & c_1 + 3a_1 & 2a_1 + 3b_1 \\ 2b_2 + c_2 & c_2 + 3a_2 & 2a_2 + 3b_2 \\ 2b_3 + c_3 & c_3 + 3a_3 & 2a_3 + 3b_3 \end{vmatrix} \text{ is a multiple of } \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

**Solution:**

Express the given matrix as a product of two matrices:

$$\begin{vmatrix} 2b_1 + c_1 & c_1 + 3a_1 & 2a_1 + 3b_1 \\ 2b_2 + c_2 & c_2 + 3a_2 & 2a_2 + 3b_2 \\ 2b_3 + c_3 & c_3 + 3a_3 & 2a_3 + 3b_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \times \begin{vmatrix} 0 & 3 & 2 \\ 2 & 0 & 3 \\ 1 & 1 & 0 \end{vmatrix}$$

Calculate the determinant of the secondary matrix:

$$\begin{vmatrix} 0 & 3 & 2 \\ 2 & 0 & 3 \\ 1 & 1 & 0 \end{vmatrix} = 0(0-3) - 3(0-3) + 2(2-0) = 0 + 9 + 4 = 13$$

Since the determinant of a product is the product of the determinants ( $\det(AB) = \det(A) \det(B)$ ):

The original determinant is exactly **13** times  $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ .

**Multiplier: 13.**

**Q3. Prove the squared differences identity.**

$$\begin{vmatrix} (a-x)^2 & (a-y)^2 & (a-z)^2 \\ (b-x)^2 & (b-y)^2 & (b-z)^2 \\ (c-x)^2 & (c-y)^2 & (c-z)^2 \end{vmatrix} = 2(a-b)(b-c)(c-a)(x-y)(y-z)(z-x)$$

**Solution:**

Expanding the identity  $(a-x)^2 = a^2 - 2ax + x^2$ , the matrix can be factored into a product of two matrices:

$$\text{L.H.S} = \begin{vmatrix} a^2 & -2a & 1 \\ b^2 & -2b & 1 \\ c^2 & -2c & 1 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

Evaluate the first determinant by factoring out  $-2$  from the second column and swapping columns ( $C_1 \leftrightarrow C_3$ ) to match the standard Vandermonde format (swapping columns negates the determinant):

$$(-2)(-1) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = 2(a-b)(b-c)(c-a)$$

Evaluate the second determinant (Standard Vandermonde Matrix):

$$\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^2 & y^2 & z^2 \end{vmatrix} = (x-y)(y-z)(z-x)$$

Multiply both results together:

$$\text{L.H.S} = 2(a-b)(b-c)(c-a)(x-y)(y-z)(z-x) = \text{R.H.S} \quad (\text{Proved})$$

**Q4. Evaluate the Determinant of the 5th Roots of Unity.** Evaluate the determinant of matrix  $M$  where  $\omega$  is the complex 5th root of unity ( $\omega^5 = 1$ ):

$$M = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & \omega & \omega^2 & \omega^3 & \omega^4 \\ 1 & \omega^2 & \omega^4 & \omega^6 & \omega^8 \\ 1 & \omega^3 & \omega^6 & \omega^9 & \omega^{12} \\ 1 & \omega^4 & \omega^8 & \omega^{12} & \omega^{16} \end{bmatrix}$$

**Solution:**

By multiplying matrix  $M$  by itself ( $M \times M$ ), we leverage the property of roots of unity:  $1 + \omega^k + \omega^{2k} + \omega^{3k} + \omega^{4k} = 0$  (for  $k$  not a multiple of 5) and equals 5 when  $k$  is a multiple of 5.

$$M^2 = \begin{bmatrix} 5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 5 & 0 \\ 0 & 0 & 5 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 \end{bmatrix}$$

Evaluate the determinant of the resulting anti-diagonal block matrix  $M^2$ :

$$\det(M^2) = 5 \begin{vmatrix} 0 & 0 & 0 & 5 \\ 0 & 0 & 5 & 0 \\ 0 & 5 & 0 & 0 \\ 5 & 0 & 0 & 0 \end{vmatrix} = 5 \times (5^4) = 3125$$

Therefore, the square of the determinant is **3125** (or  $5^5$ ).

**Q5. Prove the Symmetric Matrix Product Identity.**

$$\begin{vmatrix} b^2 + c^2 & ab & ac \\ ab & c^2 + a^2 & bc \\ ac & bc & a^2 + b^2 \end{vmatrix} = 4a^2b^2c^2$$

**Solution:**

Recognize that this complex matrix is simply the product of a simpler matrix multiplied by itself (or its transpose):

$$\text{Let } A = \begin{vmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{vmatrix}$$

$$A \times A = \begin{bmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{bmatrix} \begin{bmatrix} 0 & c & b \\ c & 0 & a \\ b & a & 0 \end{bmatrix} = \begin{bmatrix} c^2 + b^2 & ab & ac \\ ab & c^2 + a^2 & bc \\ ac & bc & a^2 + b^2 \end{bmatrix}$$

Calculate  $\det(A)$  by expanding along the first row:

$$\det(A) = 0 - c(0 - ab) + b(ca - 0) = abc + abc = 2abc$$

The determinant of the product is simply  $\det(A) \times \det(A)$ :

$$\text{L.H.S} = (2abc)^2 = \mathbf{4a^2b^2c^2} = \text{R.H.S} \quad (\text{Proved})$$

## Lecture Questions: Matrix Operations & Applications

Advanced matrix applications: Step-by-step Echelon forms, Rank, and Inverses.

**Q1. Solve the following system of linear equations using Echelon Form (Gaussian Elimination).**

$$\begin{aligned} 2x_1 + x_3 &= 1 \\ 2x_1 + 4x_2 - x_3 &= -2 \\ x_1 - 8x_2 - 3x_3 &= 2 \end{aligned}$$

**Solution:**

Set up the augmented matrix  $[A \mid B]$ :

$$\begin{bmatrix} 2 & 0 & 1 & 1 \\ 2 & 4 & -1 & -2 \\ 1 & -8 & -3 & 2 \end{bmatrix} R_1 \leftrightarrow R_3 \sim \begin{bmatrix} 1 & -8 & -3 & 2 \\ 2 & 4 & -1 & -2 \\ 2 & 0 & 1 & 1 \end{bmatrix}$$

Eliminate entries below the first pivot:

$$\begin{aligned} R_2 &\rightarrow R_2 - 2R_1 \\ R_3 &\rightarrow R_3 - 2R_1 \end{aligned} \sim \begin{bmatrix} 1 & -8 & -3 & 2 \\ 0 & 20 & 5 & -6 \\ 0 & 16 & 7 & -3 \end{bmatrix}$$

Make the second pivot exactly 1:

$$R_2 \rightarrow \frac{1}{20}R_2 \sim \begin{bmatrix} 1 & -8 & -3 & 2 \\ 0 & 1 & 1/4 & -3/10 \\ 0 & 16 & 7 & -3 \end{bmatrix}$$

Eliminate entries below the second pivot:

$$R_3 \rightarrow R_3 - 16R_2 \sim \begin{bmatrix} 1 & -8 & -3 & 2 \\ 0 & 1 & 1/4 & -3/10 \\ 0 & 0 & 3 & 9/5 \end{bmatrix}$$

Make the third pivot exactly 1:

$$R_3 \rightarrow \frac{1}{3}R_3 \sim \begin{bmatrix} 1 & -8 & -3 & 2 \\ 0 & 1 & 1/4 & -3/10 \\ 0 & 0 & 1 & 3/5 \end{bmatrix}$$

The matrix is now in strict Row Echelon Form with 1s on the diagonal. Applying backward substitution:

From  $R_3$ :  $\mathbf{x}_3 = \frac{3}{5}$

From  $R_2$ :  $x_2 + \frac{1}{4}(\frac{3}{5}) = -\frac{3}{10} \implies x_2 + \frac{3}{20} = -\frac{6}{20} \implies \mathbf{x}_2 = -\frac{9}{20}$

From  $R_1$ :  $x_1 - 8(-\frac{9}{20}) - 3(\frac{3}{5}) = 2 \implies x_1 + \frac{18}{5} - \frac{9}{5} = 2 \implies \mathbf{x}_1 = \frac{1}{5}$

**Q2. Solve the following system of equations using the Gauss-Jordan Method (Reduced Row Echelon Form).**

$$\begin{aligned} 2x_1 + x_2 - x_3 &= 3 \\ x_1 + 3x_2 - 5x_3 &= 2 \\ 4x_1 - x_2 + x_3 &= 3 \end{aligned}$$

**Solution:**

Set up the augmented matrix  $[A | B]$  and apply row operations to reach RREF:

$$\begin{bmatrix} 2 & 1 & -1 & 3 \\ 1 & 3 & -5 & 2 \\ 4 & -1 & 1 & 3 \end{bmatrix} R_1 \leftrightarrow R_2 \sim \begin{bmatrix} 1 & 3 & -5 & 2 \\ 2 & 1 & -1 & 3 \\ 4 & -1 & 1 & 3 \end{bmatrix}$$

Zero out Column 1:

$$\begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 4R_1 \end{matrix} \sim \begin{bmatrix} 1 & 3 & -5 & 2 \\ 0 & -5 & 9 & -1 \\ 0 & -13 & 21 & -5 \end{bmatrix}$$

Make Pivot 2 exactly 1:

$$R_2 \rightarrow -\frac{1}{5}R_2 \sim \begin{bmatrix} 1 & 3 & -5 & 2 \\ 0 & 1 & -9/5 & 1/5 \\ 0 & -13 & 21 & -5 \end{bmatrix}$$

Zero out below Pivot 2:

$$R_3 \rightarrow R_3 + 13R_2 \sim \begin{bmatrix} 1 & 3 & -5 & 2 \\ 0 & 1 & -9/5 & 1/5 \\ 0 & 0 & -12/5 & -12/5 \end{bmatrix}$$

Make Pivot 3 exactly 1:

$$R_3 \rightarrow -\frac{5}{12}R_3 \sim \begin{bmatrix} 1 & 3 & -5 & 2 \\ 0 & 1 & -9/5 & 1/5 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Now eliminate upwards to reach RREF (zeroing out above pivots):

$$\begin{matrix} R_2 \rightarrow R_2 + \frac{9}{5}R_3 \\ R_1 \rightarrow R_1 + 5R_3 \end{matrix} \sim \begin{bmatrix} 1 & 3 & 0 & 7 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 3R_2 \sim \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Extracting the values directly from the resulting identity matrix: **Answer:  $x_1 = 1, x_2 = 2, x_3 = 1$ .**

**Q3. Solve the following  $4 \times 4$  system of linear equations using Echelon Form.**

$$\begin{aligned} x_1 - 2x_2 - 7x_3 + x_4 &= 5 \\ -x_1 + 2x_2 + 8x_3 - 5x_4 &= -7 \\ 3x_1 - 4x_2 - 17x_3 + 13x_4 &= 14 \\ 2x_1 - 2x_2 - 11x_3 + 2x_4 &= 7 \end{aligned}$$

**Solution:**

Set up the augmented matrix  $[A | B]$ :

$$\left[ \begin{array}{cccc|c} 1 & -2 & -7 & 1 & 5 \\ -1 & 2 & 8 & -5 & -7 \\ 3 & -4 & -17 & 13 & 14 \\ 2 & -2 & -11 & 2 & 7 \end{array} \right] \begin{matrix} R_2 \rightarrow R_2 + R_1 \\ R_3 \rightarrow R_3 - 3R_1 \\ R_4 \rightarrow R_4 - 2R_1 \end{matrix} \sim \left[ \begin{array}{cccc|c} 1 & -2 & -7 & 1 & 5 \\ 0 & 0 & 1 & -4 & -2 \\ 0 & 2 & 4 & 10 & -1 \\ 0 & 2 & 3 & 0 & -3 \end{array} \right]$$

Swap rows to position a non-zero pivot in the second column:

$$R_2 \leftrightarrow R_3 \sim \begin{bmatrix} 1 & -2 & -7 & 1 & | & 5 \\ 0 & 2 & 4 & 10 & | & -1 \\ 0 & 0 & 1 & -4 & | & -2 \\ 0 & 2 & 3 & 0 & | & -3 \end{bmatrix}$$

Make the second pivot exactly 1:

$$R_2 \rightarrow \frac{1}{2}R_2 \sim \begin{bmatrix} 1 & -2 & -7 & 1 & | & 5 \\ 0 & 1 & 2 & 5 & | & -1/2 \\ 0 & 0 & 1 & -4 & | & -2 \\ 0 & 2 & 3 & 0 & | & -3 \end{bmatrix}$$

Eliminate below the second pivot:

$$R_4 \rightarrow R_4 - 2R_2 \sim \begin{bmatrix} 1 & -2 & -7 & 1 & | & 5 \\ 0 & 1 & 2 & 5 & | & -1/2 \\ 0 & 0 & 1 & -4 & | & -2 \\ 0 & 0 & -1 & -10 & | & -2 \end{bmatrix}$$

Eliminate below the third pivot:

$$R_4 \rightarrow R_4 + R_3 \sim \begin{bmatrix} 1 & -2 & -7 & 1 & | & 5 \\ 0 & 1 & 2 & 5 & | & -1/2 \\ 0 & 0 & 1 & -4 & | & -2 \\ 0 & 0 & 0 & -14 & | & -4 \end{bmatrix}$$

Make the fourth pivot exactly 1:

$$R_4 \rightarrow -\frac{1}{14}R_4 \sim \begin{bmatrix} 1 & -2 & -7 & 1 & | & 5 \\ 0 & 1 & 2 & 5 & | & -1/2 \\ 0 & 0 & 1 & -4 & | & -2 \\ 0 & 0 & 0 & 1 & | & 2/7 \end{bmatrix}$$

Applying backward substitution:

From  $R_4$ :  $\mathbf{x}_4 = \frac{2}{7}$

From  $R_3$ :  $x_3 - 4(\frac{2}{7}) = -2 \implies x_3 - \frac{8}{7} = -2 \implies \mathbf{x}_3 = -\frac{6}{7}$

From  $R_2$ :  $x_2 + 2(-\frac{6}{7}) + 5(\frac{2}{7}) = -\frac{1}{2} \implies x_2 - \frac{12}{7} + \frac{10}{7} = -\frac{1}{2} \implies \mathbf{x}_2 = -\frac{3}{14}$

From  $R_1$ :  $x_1 - 2(-\frac{3}{14}) - 7(-\frac{6}{7}) + \frac{2}{7} = 5 \implies x_1 + \frac{3}{7} + 6 + \frac{2}{7} = 5 \implies \mathbf{x}_1 = -\frac{12}{7}$

**Q4. Find the inverse of the matrix using the Gauss-Jordan Method.**

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$$

**Solution:**

Set up the augmented matrix  $[A | I]$  and reduce the left side to  $I$ :

$$\begin{bmatrix} 2 & 1 & -1 & | & 1 & 0 & 0 \\ 0 & 2 & 1 & | & 0 & 1 & 0 \\ 5 & 2 & -3 & | & 0 & 0 & 1 \end{bmatrix} R_1 \rightarrow \frac{1}{2}R_1 \sim \begin{bmatrix} 1 & 1/2 & -1/2 & | & 1/2 & 0 & 0 \\ 0 & 2 & 1 & | & 0 & 1 & 0 \\ 5 & 2 & -3 & | & 0 & 0 & 1 \end{bmatrix}$$



$$R_3 \rightarrow R_3 - 5R_1 \sim \begin{bmatrix} 1 & 1/2 & -1/2 & | & 1/2 & 0 & 0 \\ 0 & 2 & 1 & | & 0 & 1 & 0 \\ 0 & -1/2 & -1/2 & | & -5/2 & 0 & 1 \end{bmatrix}$$

$$R_2 \rightarrow \frac{1}{2}R_2 \sim \begin{bmatrix} 1 & 1/2 & -1/2 & | & 1/2 & 0 & 0 \\ 0 & 1 & 1/2 & | & 0 & 1/2 & 0 \\ 0 & -1/2 & -1/2 & | & -5/2 & 0 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 + \frac{1}{2}R_2 \sim \begin{bmatrix} 1 & 1/2 & -1/2 & | & 1/2 & 0 & 0 \\ 0 & 1 & 1/2 & | & 0 & 1/2 & 0 \\ 0 & 0 & -1/4 & | & -5/2 & 1/4 & 1 \end{bmatrix}$$

$$R_3 \rightarrow -4R_3 \sim \begin{bmatrix} 1 & 1/2 & -1/2 & | & 1/2 & 0 & 0 \\ 0 & 1 & 1/2 & | & 0 & 1/2 & 0 \\ 0 & 0 & 1 & | & 10 & -1 & -4 \end{bmatrix}$$

Eliminate upwards:

$$\begin{matrix} R_2 \rightarrow R_2 - \frac{1}{2}R_3 \\ R_1 \rightarrow R_1 + \frac{1}{2}R_3 \end{matrix} \sim \begin{bmatrix} 1 & 1/2 & 0 & | & 11/2 & -1/2 & -2 \\ 0 & 1 & 0 & | & -5 & 1 & 2 \\ 0 & 0 & 1 & | & 10 & -1 & -4 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - \frac{1}{2}R_2 \sim \begin{bmatrix} 1 & 0 & 0 & | & 8 & -1 & -3 \\ 0 & 1 & 0 & | & -5 & 1 & 2 \\ 0 & 0 & 1 & | & 10 & -1 & -4 \end{bmatrix}$$

**Answer:**  $A^{-1} = \begin{bmatrix} 8 & -1 & -3 \\ -5 & 1 & 2 \\ 10 & -1 & -4 \end{bmatrix}$

**Q5. Find the Rank of the given matrix.**

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 1 & 0 \\ -2 & -1 & 3 \\ -1 & 4 & -2 \end{bmatrix}$$

**Solution:**

To find the rank, convert the matrix to Row Echelon Form with leading 1s.

$$\begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 + 2R_1 \\ R_4 \rightarrow R_4 + R_1 \end{matrix} \sim \begin{bmatrix} 1 & 2 & -3 \\ 0 & -3 & 6 \\ 0 & 3 & -3 \\ 0 & 6 & -5 \end{bmatrix}$$

Make pivot 2 a 1:

$$R_2 \rightarrow -\frac{1}{3}R_2 \sim \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & -2 \\ 0 & 3 & -3 \\ 0 & 6 & -5 \end{bmatrix}$$

Eliminate below pivot 2:

$$\begin{matrix} R_3 \rightarrow R_3 - 3R_2 \\ R_4 \rightarrow R_4 - 6R_2 \end{matrix} \sim \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & -2 \\ 0 & 0 & 3 \\ 0 & 0 & 7 \end{bmatrix}$$

Make pivot 3 a 1:

$$R_3 \rightarrow \frac{1}{3}R_3 \sim \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \\ 0 & 0 & 7 \end{bmatrix}$$

Eliminate below pivot 3:

$$R_4 \rightarrow R_4 - 7R_3 \sim \begin{bmatrix} 1 & 2 & -3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

The number of non-zero rows after converting the matrix into strict echelon form is the rank. Since there are 3 non-zero rows, the **Rank of the matrix is 3**.

## Chapter 4: General Vector Spaces

### Key Concepts to Know, Revise, and Understand

Before solving the exercises in 4.2 and 4.3, you must have a solid grasp of how vector spaces behave, how vectors combine, and how to test for independence.

#### Subspaces

A subset  $W$  of a vector space  $V$  is called a **subspace** if  $W$  is itself a vector space under the addition and scalar multiplication defined on  $V$ . To prove  $W$  is a subspace, you only need to verify three axioms:

1. **The Zero Vector:** The zero vector  $\mathbf{0}$  of  $V$  must be in  $W$ .
2. **Closure under Addition:** If  $\mathbf{u}$  and  $\mathbf{v}$  are in  $W$ , then  $\mathbf{u} + \mathbf{v}$  must also be in  $W$ .
3. **Closure under Scalar Multiplication:** If  $\mathbf{u}$  is in  $W$  and  $k$  is any scalar, then  $k\mathbf{u}$  must be in  $W$ .

#### Pro-Tip / Core Intuition

##### Zero Vector Test

Always test the zero vector first! If setting the parameters to zero doesn't yield the exact zero vector for that space, it immediately fails the first axiom and **is not a subspace**. This saves you from having to prove closure.

#### Linear Combinations & Spanning Sets

##### Linear Combinations:

A vector  $\mathbf{w}$  is a **linear combination** of the vectors  $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$  if it can be expressed as:

$$\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n$$

where  $c_1, c_2, \dots, c_n$  are scalars. If the resulting system of equations is inconsistent, it is not a linear combination.

##### Spanning Sets:

If *every* vector in a vector space  $V$  can be expressed as a linear combination of the vectors in a set  $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ , then  $S$  **spans**  $V$ .

### Linear Independence vs. Dependence

A set of vectors  $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$  is **linearly independent** if the vector equation:

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n = \mathbf{0}$$

has *only* the trivial solution ( $c_1 = 0, c_2 = 0, \dots, c_n = 0$ ). If there are other (non-trivial) solutions, the set is **linearly dependent**.

- **Determinant Test (Square Matrices):** Form a matrix. If  $\det(A) \neq 0$ , the vectors are linearly independent. If  $\det(A) = 0$ , they are dependent.
- **Rank Test (Rectangular Matrices):** Reduce to echelon form. If the rank (number of non-zero rows) equals the number of vectors, they are linearly independent.

### Why This Matters: Real-World Applications

Why Linear Independence Matters in Tech

- **Data Compression (Machine Learning):** Linearly dependent data points carry redundant, overlapping information. Algorithms like PCA actively remove dependent features to compress datasets (like images) without losing critical data.
- **Computer Graphics:** To render a 3D environment, your engine needs exactly 3 linearly independent basis vectors ( $x, y, z$ ). If they become dependent (a phenomenon known as *Gimbal lock*), two axes align, and you completely lose a dimension of rotation!

### The Wronskian

To check if a set of functions  $f_1(x), f_2(x), \dots, f_n(x)$  is linearly independent, we use the Wronskian determinant:

$$W(x) = \begin{vmatrix} f_1 & f_2 & \dots & f_n \\ f_1' & f_2' & \dots & f_n' \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)} & f_2^{(n-1)} & \dots & f_n^{(n-1)} \end{vmatrix}$$

If  $W(x) \neq 0$  for at least one value of  $x$ , the functions are linearly independent.

## Detailed Solutions to Exercises 4.2 & 4.3

### Exercise 4.2

#### Question 1

**Problem:** Use theorem 4.2.1 to determine which of the following are subspaces of  $\mathbb{R}^3$  or  $M_{nn}$ .

**(d)** All vectors of the form  $(a, b, c)$ , where  $b = a + c + 1$ .

**Solution:**

For a set to be a subspace, it must contain the zero vector  $(0, 0, 0)$ . Substituting  $a = 0$  and  $c = 0$  into the given condition  $b = a + c + 1$ :

$$b = 0 + 0 + 1 = 1$$

This yields the vector  $(0, 1, 0)$ , not  $(0, 0, 0)$ . Since the zero vector  $(0, 0, 0)$  is not in the set, it fails the vector space axioms. **It is not a subspace.**

**(e)** Determine whether the set of all  $n \times n$  matrices  $A$  such that  $A^T = -A$  is a subspace of  $M_{nn}$ .

**Solution:**

i) Closure under Addition: Let  $A$  and  $B$  be in the set.

$$(A + B)^T = A^T + B^T$$

Since they are skew-symmetric, substitute  $A^T = -A$  and  $B^T = -B$ :

$$(A + B)^T = -A - B = -(A + B)$$

The sum is also in the set.

ii) Closure under Scalar Multiplication: Let  $k$  be a scalar.

$$(kA)^T = k(A^T) = k(-A) = -(kA)$$

The scalar multiple is also in the set. **It is a subspace.**

---

#### Question 7

**Problem:** Which of the following are linear combinations of  $\mathbf{u} = (0, -2, 2)$  and  $\mathbf{v} = (1, 3, -1)$ ?

(d)  $(0, 4, 5)$

**Solution:**

If  $(0, 4, 5)$  is a linear combination, it can be written as:

$$a(0, -2, 2) + b(1, 3, -1) = (0, 4, 5)$$

This forms the following system of equations:

$$0a + b = 0 \implies b = 0 \quad (1)$$

$$-2a + 3b = 4 \quad (2)$$

$$2a - b = 5 \quad (3)$$

Substitute  $b = 0$  into equation (2):

$$-2a = 4$$

$$a = -2$$

Verify with equation (3):

$$2(-2) - 0 = -4$$

$$-4 \neq 5$$

The system is inconsistent.  $(0, 4, 5)$  **is not a linear combination.**

## Question 8

**Problem:** Express the following as linear combinations of  $\mathbf{u} = (2, 1, 4)$ ,  $\mathbf{v} = (1, -1, 3)$ , and  $\mathbf{w} = (3, 2, 5)$ .

(a)  $(-9, -7, -15)$

**Solution:**

Set up the linear combination of scalars:

$$a(2, 1, 4) + b(1, -1, 3) + c(3, 2, 5) = (-9, -7, -15)$$

By equating the components, we get:

$$2a + b + 3c = -9 \quad (1)$$

$$a - b + 2c = -7 \quad (2)$$

$$4a + 3b + 5c = -15 \quad (3)$$

Adding eq(1) and eq(2):

$$\begin{array}{rcl} 2a + b + 3c & = & -9 \\ + \quad a - b + 2c & = & -7 \\ \hline 3a + 5c & = & -16 \end{array} \quad (4)$$

Multiplying eq(2) by 3 and adding this new eq to eq(3):

$$\begin{array}{r} 4a + 3b + 5c = -15 \\ + \quad 3a - 3b + 6c = -21 \\ \hline 7a + 11c = -36 \end{array} \quad (5)$$

From eq(4), isolating  $c$ :

$$\begin{array}{r} 5c = -16 - 3a \\ c = \frac{-16 - 3a}{5} \end{array} \quad (6)$$

Substituting this  $c$  into eq(5):

$$\begin{aligned} 7a + 11 \left( \frac{-16 - 3a}{5} \right) &= -36 \\ 35a + 11(-16 - 3a) &= -180 \\ 35a - 176 - 33a &= -180 \\ 2a - 176 &= -180 \\ 2a &= -4 \\ a &= -2 \end{aligned}$$

Substituting  $a = -2$  back into eq(4):

$$\begin{aligned} 3(-2) + 5c &= -16 \\ -6 + 5c &= -16 \\ 5c &= -10 \\ c &= -2 \end{aligned}$$

Substituting  $a = -2$  and  $c = -2$  into eq(2):

$$\begin{aligned} (-2) - b + 2(-2) &= -7 \\ -2 - b - 4 &= -7 \\ -6 &= -7 + b \\ b &= 1 \end{aligned}$$

Final expression:

$$(-9, -7, -15) = -2\mathbf{u} + 1\mathbf{v} - 2\mathbf{w}$$

## Question 10

**Problem:** Express the vector as a linear combination of  $\mathbf{p}_1 = 2 + x + 4x^2$ ,  $\mathbf{p}_2 = 1 - x + 3x^2$ , and  $\mathbf{p}_3 = 3 + 2x + 5x^2$ .

**(a)**  $6 + 11x + 6x^2$

**Solution:**

Set up the linear combination of scalars:

$$a(2 + x + 4x^2) + b(1 - x + 3x^2) + c(3 + 2x + 5x^2) = 6 + 11x + 6x^2$$

This forms the following system of equations:

$$\text{(constants)} \quad 2a + b + 3c = 6 \quad (1)$$

$$\text{(x terms)} \quad a - b + 2c = 11 \quad (2)$$

$$\text{(x}^2 \text{ terms)} \quad 4a + 3b + 5c = 6 \quad (3)$$

Adding eq(1) and eq(2):

$$\begin{array}{r} 2a + b + 3c = 6 \\ + \quad a - b + 2c = 11 \\ \hline 3a + 5c = 17 \end{array} \quad (4)$$

Multiplying eq(2) by 3 and adding this new eq to eq(3):

$$\begin{array}{r} 3a - 3b + 6c = 33 \\ + \quad 4a + 3b + 5c = 6 \\ \hline 7a + 11c = 39 \end{array} \quad (5)$$

From eq(4), isolating  $c$ :

$$\begin{aligned} 5c &= 17 - 3a \\ c &= \frac{17 - 3a}{5} \end{aligned} \quad (6)$$

Substituting this  $c$  into eq(5):

$$\begin{aligned} 7a + 11 \left( \frac{17 - 3a}{5} \right) &= 39 \\ 35a + 11(17 - 3a) &= 195 \\ 35a + 187 - 33a &= 195 \\ 2a + 187 &= 195 \\ 2a &= 8 \\ a &= 4 \end{aligned}$$

Substituting  $a = 4$  back into eq(4):

$$\begin{aligned} 3(4) + 5c &= 17 \\ 12 + 5c &= 17 \\ 5c &= 5 \\ c &= 1 \end{aligned}$$

Substituting  $a = 4$  and  $c = 1$  into eq(2):

$$\begin{aligned} (4) - b + 2(1) &= 11 \\ 6 - b &= 11 \\ -b &= 5 \\ b &= -5 \end{aligned}$$



Final expression:

$$6 + 11x + 6x^2 = 4p_1 - 5p_2 + 1p_3$$

### Question 9(c)

**Problem:** Which of the following are linear combinations of  $A = \begin{bmatrix} 4 & 0 \\ -2 & -2 \end{bmatrix}$ ,  $B = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ ,  $C = \begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix}$ ?

(c)  $\begin{bmatrix} -1 & 5 \\ 7 & 1 \end{bmatrix}$

**Solution:**

Set up the linear combination of scalars:

$$a \begin{bmatrix} 4 & 0 \\ -2 & -2 \end{bmatrix} + b \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} + c \begin{bmatrix} 0 & 2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 7 & 1 \end{bmatrix}$$

This forms the following system of equations:

$$4a + b = -1 \quad (1)$$

$$-b + 2c = 5 \quad (2)$$

$$-2a + 2b + c = 7 \quad (3)$$

$$-2a + 3b + 4c = 1 \quad (4)$$

From eq(1), isolating  $b$ :

$$b = -1 - 4a \quad (5)$$

Substituting  $b$  into eq(2):

$$-(-1 - 4a) + 2c = 5$$

$$1 + 4a + 2c = 5$$

$$2c = 4 - 4a$$

$$c = 2 - 2a \quad (6)$$

Substituting  $b$  and  $c$  into eq(3):

$$-2a + 2(-1 - 4a) + (2 - 2a) = 7$$

$$-2a - 2 - 8a + 2 - 2a = 7$$

$$-12a = 7$$

$$a = -\frac{7}{12}$$

Putting the value of  $a = -\frac{7}{12}$  into eq(5) and eq(6):

$$b = -1 - 4 \left( -\frac{7}{12} \right) = -1 + \frac{28}{12} = -1 + \frac{7}{3} = \frac{4}{3}$$

$$c = 2 - 2 \left( -\frac{7}{12} \right) = 2 + \frac{14}{12} = 2 + \frac{7}{6} = \frac{19}{6}$$

Now, substituting  $a, b,$  and  $c$  into eq(4) to check for consistency:

$$\begin{aligned} -2a + 3b + 4c &= 1 \\ -2 \left( -\frac{7}{12} \right) + 3 \left( \frac{4}{3} \right) + 4 \left( \frac{19}{6} \right) &= 1 \\ \frac{14}{12} + 4 + \frac{76}{6} &= 1 \\ \frac{7}{6} + \frac{24}{6} + \frac{76}{6} &= 1 \\ \frac{107}{6} &\neq 1 \end{aligned}$$

Because the L.H.S  $\neq$  R.H.S, the equations contradict each other. The system is inconsistent and the scalars do not exist. **The matrix is not a linear combination of A, B, and C.**

### Question 11

**Problem:** In each part, determine whether the vectors span  $\mathbb{R}^3$ .

**(a)**  $\mathbf{v}_1 = (2, 2, 2), \mathbf{v}_2 = (0, 0, 3), \mathbf{v}_3 = (0, 1, 1)$

**Solution:**

Setting up the determinant:

$$\begin{aligned} |A| &= \begin{vmatrix} 2 & 0 & 0 \\ 2 & 0 & 1 \\ 2 & 3 & 1 \end{vmatrix} \\ &= 2 \begin{vmatrix} 0 & 1 \\ 3 & 1 \end{vmatrix} = 2(0 - 3) = -6 \end{aligned}$$

Since the scalar value of the determinant is not equal to zero, the vectors are linearly independent. They can also form a basis, meaning **they span  $\mathbb{R}^3$ .**

**(b)**  $\mathbf{v}_1 = (2, -1, 3), \mathbf{v}_2 = (4, 1, 2), \mathbf{v}_3 = (8, -1, 8)$

**Solution:**

$$\begin{aligned}
 |A| &= \begin{vmatrix} 2 & 4 & 8 \\ -1 & 1 & -1 \\ 3 & 2 & 8 \end{vmatrix} \\
 &= 2(8 - (-2)) - 4(-8 - (-3)) + 8(-2 - 3) \\
 &= 2(10) - 4(-5) + 8(-5) \\
 &= 20 + 20 - 40 \\
 &= 0
 \end{aligned}$$

Since the scalar value of the determinant is equal to 0, the vectors are linearly dependent, meaning **they do not span**  $\mathbb{R}^3$ .

## Question 12

**Problem:** Suppose that  $\mathbf{v}_1 = (2, 1, 0, 3)$ ,  $\mathbf{v}_2 = (3, -1, 5, 2)$ , and  $\mathbf{v}_3 = (-1, 0, 2, 1)$ . Which of the following vectors are in  $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ ?

(a)  $(2, 3, -7, 3)$

**Solution:**

Set up the linear combination of scalars:

$$a(2, 1, 0, 3) + b(3, -1, 5, 2) + c(-1, 0, 2, 1) = (2, 3, -7, 3)$$

This forms the following system of equations:

$$2a + 3b - c = 2 \quad (1)$$

$$a - b = 3 \quad (2)$$

$$5b + 2c = -7 \quad (3)$$

$$3a + 2b + c = 3 \quad (4)$$

From eq(2), isolating  $a$ :

$$a = b + 3 \quad (5)$$

Substitute the expression for  $a$  into eq(1):

$$2(b + 3) + 3b - c = 2$$

$$2b + 6 + 3b - c = 2$$

$$5b - c = -4$$

$$c = 5b + 4 \quad (6)$$

Substituting  $c$  into eq(3):

$$5b + 2(5b + 4) = -7$$

$$5b + 10b + 8 = -7$$

$$15b = -15$$

$$b = -1$$

Putting the value of  $b = -1$  into eq(5) and eq(6):

$$a = -1 + 3 \implies a = 2$$

$$c = 5(-1) + 4 \implies c = -1$$

Verifying with the final equation (eq 4): Substitute  $a = 2, b = -1, c = -1$  into eq(4):

$$\begin{aligned} 3(2) + 2(-1) + (-1) &= 6 - 2 - 1 = 3 \\ 3 &= 3 \end{aligned}$$

Since L.H.S = R.H.S, the system is consistent. It is a linear combination, therefore **it is in the span.**

**(b)**  $(-4, 6, -13, 4)$

**Solution:**

Set up the linear combination of scalars:

$$a(2, 1, 0, 3) + b(3, -1, 5, 2) + c(-1, 0, 2, 1) = (-4, 6, -13, 4)$$

This forms the following system of equations:

$$2a + 3b - c = -4 \quad (1)$$

$$a - b = 6 \quad (2)$$

$$5b + 2c = -13 \quad (3)$$

$$3a + 2b + c = 4 \quad (4)$$

From eq(2), isolating  $a$ :

$$a = b + 6 \quad (5)$$

Putting the value of  $a$  into eq(1):

$$2(b + 6) + 3b - c = -4$$

$$2b + 12 + 3b - c = -4$$

$$5b - c = -16$$

$$c = 5b + 16 \quad (6)$$

Putting the value of  $c$  into eq(3):

$$5b + 2(5b + 16) = -13$$

$$15b + 32 = -13$$

$$15b = -45$$

$$b = -3$$

Putting the value of  $b = -3$  into eq(5) and eq(6):

$$a = (-3) + 6 \implies a = 3$$

$$c = 5(-3) + 16 \implies c = 1$$

Putting the values  $a = 3, b = -3, c = 1$  into eq(4) to verify:

$$3(3) + 2(-3) + (1) = 9 - 6 + 1 = 4$$

Since L.H.S = R.H.S, the system is consistent. It is a linear combination, therefore **it is in the span.**

---

## Exercise 4.3

### Question 2

**Problem:** In each part, determine whether the vectors are linearly dependent or independent in  $\mathbb{R}^3$ .

**(a)**  $(-3, 0, 4), (5, -1, 2), (1, 1, 3)$

**Solution:**

Because we have 3 vectors in  $\mathbb{R}^3$ , we can form a square matrix and use the determinant method to check for linear independence.

$$\begin{aligned}\det(A) &= \begin{vmatrix} -3 & 0 & 4 \\ 5 & -1 & 2 \\ 1 & 1 & 3 \end{vmatrix} \\ &= -3(-3 - 2) + 4(5 - (-1)) \\ &= -3(-5) + 4(6) \\ &= 15 + 24 = 39\end{aligned}$$

Since  $\det(A) = 39 \neq 0$ , **the vectors are linearly independent.**

---

### Question 3

**Problem:** In each part, determine whether the vectors are linearly independent or linearly dependent in  $\mathbb{R}^4$ .

**(a)**  $(3, 0, -3, 6), (0, 2, 3, 1), (0, -2, -2, 0), (-2, 1, 2, 1)$

**Solution:**

Because we have 4 vectors in  $\mathbb{R}^4$ , we can stack them as rows to form a matrix and use the

rank method to check for linear independence.

$$A = \begin{bmatrix} 3 & 0 & -3 & 6 \\ 0 & 2 & 3 & 1 \\ 0 & -2 & -2 & 0 \\ -2 & 1 & 2 & 1 \end{bmatrix}$$

$$\begin{array}{l} R_1 \rightarrow R_1 \div 3 \\ R_3 \rightarrow R_3 + R_2 \\ R_4 \rightarrow R_4 + 2R_1 \end{array} \sim \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 2 & 3 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 5 \end{bmatrix}$$

$$R_2 \leftrightarrow R_4 \sim \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 1 \\ 0 & 2 & 3 & 1 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - 2R_2 \sim \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 3 & -9 \end{bmatrix}$$

$$R_4 \rightarrow R_4 - 3R_3 \sim \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -12 \end{bmatrix}$$

Since the rank (4) is exactly equal to the number of vectors, **the vectors are linearly independent.**

### Question 5

**Problem:** In each part determine whether the matrices  $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ ,  $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ ,  $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$  are linearly independent or dependent in  $M_{23}$ .

**Solution:**

To check the linear independence, we set their linear combination equal to the zero matrix:

$$a \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + b \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} + c \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} a & 0 & b \\ 0 & c & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Equating the corresponding components gives:

$$a = 0, \quad b = 0, \quad c = 0$$

Since  $a = 0, b = 0, c = 0$  is the only solution (the trivial solution), **the matrices are linearly independent.**

### Question 10

**Problem:** Show that the vectors  $\mathbf{v}_1 = (1, 2, 3, 4)$ ,  $\mathbf{v}_2 = (0, 1, 0, -1)$ , and  $\mathbf{v}_3 = (1, 3, 3, 3)$  form a linearly dependent set in  $\mathbb{R}^4$ .

**Solution:**

To check for dependence, we set their linear combination equal to the zero vector:

$$a(1, 2, 3, 4) + b(0, 1, 0, -1) + c(1, 3, 3, 3) = (0, 0, 0, 0)$$

This forms the following system of equations:

$$a + c = 0 \quad (1)$$

$$2a + b + 3c = 0 \quad (2)$$

$$3a + 3c = 0 \quad (3)$$

$$4a - b + 3c = 0 \quad (4)$$

From eq(1), we get  $a = -c$ . Put  $a = -c$  in eq(2):

$$2(-c) + b + 3c = 0$$

$$-2c + b + 3c = 0$$

$$b + c = 0$$

$$b = -c$$

If we let  $c = -1$ , then  $a = 1$ ,  $b = 1$ . Since we found a non-trivial solution where the scalars  $a, b, c$  are not zero, **the vectors are linearly dependent.**

### Question 11

**Problem:** For which values of  $\lambda$  do the following vectors form a dependent set in  $\mathbb{R}^3$ ?  
 $\mathbf{v}_1 = (\lambda, -1/2, -1/2)$ ,  $\mathbf{v}_2 = (-1/2, \lambda, -1/2)$ ,  $\mathbf{v}_3 = (-1/2, -1/2, \lambda)$

**Solution:**

Because we have 3 vectors in  $\mathbb{R}^3$ , we can form a square matrix. For the vectors to be linearly dependent, their determinant must equal zero.

$$\det(A) = \begin{vmatrix} \lambda & -1/2 & -1/2 \\ -1/2 & \lambda & -1/2 \\ -1/2 & -1/2 & \lambda \end{vmatrix} = 0$$

Applying  $C_1 \rightarrow C_1 + C_2 + C_3$ :

$$\begin{vmatrix} \lambda - 1 & -1/2 & -1/2 \\ \lambda - 1 & \lambda & -1/2 \\ \lambda - 1 & -1/2 & \lambda \end{vmatrix} = 0$$

Applying  $R_2 \rightarrow R_2 - R_1$  and  $R_3 \rightarrow R_3 - R_1$ :

$$\begin{vmatrix} \lambda - 1 & -1/2 & -1/2 \\ 0 & \lambda + 1/2 & 0 \\ 0 & 0 & \lambda + 1/2 \end{vmatrix} = 0$$

Evaluating the diagonal:

$$(\lambda - 1)(\lambda + 1/2)^2 = 0$$

$$\lambda = 1, \quad \lambda = -1/2$$

**The vectors are linearly dependent when  $\lambda = 1$  and  $\lambda = -1/2$ .**

### Question 16

**Problem:** By using appropriate identities, where required, determine which of the following sets of vectors in  $F(-\infty, \infty)$  are linearly dependent.

**(d)**  $\cos 2x, \sin^2 x, \cos^2 x$

**Solution:**

To check for linear independence, we set their linear combination equal to zero:

$$a(\cos 2x) + b(\sin^2 x) + c(\cos^2 x) = 0$$

Using the standard trigonometric identity  $\cos 2x = \cos^2 x - \sin^2 x$ , we can rewrite this as:

$$1(\cos 2x) + 1(\sin^2 x) - 1(\cos^2 x) = 0$$

This matches our combination format where  $a = 1$ ,  $b = 1$ , and  $c = -1$ . Since the scalars  $a, b, c$  are not zero (there is a non-trivial solution), **the functions are linearly dependent.**

### Question 19

**Problem:** Use the Wronskian to show the sets of vectors are independent.

**(a)**  $1, x, e^x$

**Solution:**

We form a Wronskian matrix using the functions and their derivatives.

$$f_1 = 1, \quad f_2 = x, \quad f_3 = e^x$$

$$f_1' = 0, \quad f_2' = 1, \quad f_3' = e^x$$

$$f_1'' = 0, \quad f_2'' = 0, \quad f_3'' = e^x$$



$$W(x) = \begin{vmatrix} 1 & x & e^x \\ 0 & 1 & e^x \\ 0 & 0 & e^x \end{vmatrix}$$

Expanding along the first column:

$$W(x) = 1(1 \cdot e^x - 0) = e^x$$

Since  $\det(W) = e^x \neq 0$ , **the functions are linearly independent.**

**(b)**  $1, x, x^2$

**Solution:**

We form a Wronskian matrix using the functions and their derivatives.

$$f_1 = 1, \quad f_2 = x, \quad f_3 = x^2$$

$$f'_1 = 0, \quad f'_2 = 1, \quad f'_3 = 2x$$

$$f''_1 = 0, \quad f''_2 = 0, \quad f''_3 = 2$$

$$W(x) = \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & 2x \\ 0 & 0 & 2 \end{vmatrix}$$

Expanding along the first column:

$$W(x) = 1(2 - 0) - 0 + 0 = 2$$

Since  $\det(W) = 2 \neq 0$ , **the functions are linearly independent.**

## Question 21

**Problem:** Use the Wronskian to show that the functions  $f_1(x) = \sin x$ ,  $f_2(x) = \cos x$ , and  $f_3(x) = x \cos x$  are linearly independent vectors in  $C^\infty(-\infty, \infty)$ .

**Solution:**

$$W(x) = \begin{vmatrix} f_1 & f_2 & f_3 \\ f'_1 & f'_2 & f'_3 \\ f''_1 & f''_2 & f''_3 \end{vmatrix}$$

$$W(x) = \begin{vmatrix} \sin x & \cos x & x \cos x \\ \cos x & -\sin x & \cos x - x \sin x \\ -\sin x & -\cos x & -2 \sin x - x \cos x \end{vmatrix}$$

Simplify using row operations:

$$W(x) \xrightarrow{R_3 \rightarrow R_3 + R_1} \begin{vmatrix} \sin x & \cos x & x \cos x \\ \cos x & -\sin x & \cos x - x \sin x \\ 0 & 0 & -2 \sin x \end{vmatrix}$$

Expanding along the third row:

$$\begin{aligned} W(x) &= -2 \sin x (-\sin^2 x - \cos^2 x) \\ &= -2 \sin x (-1) \\ &= 2 \sin x \end{aligned}$$

Since  $2 \sin x$  is not identically zero (e.g., at  $x = \pi/2$ ,  $W = 2$ ), **the functions are linearly independent.**

## Chapter 5: Eigenvalues and Eigenvectors

### Key Concepts to Know, Revise, and Understand

Before diving into the exercises, it is crucial to understand the fundamental definitions and procedures for finding eigenvalues and eigenvectors.

#### What are Eigenvalues and Eigenvectors?

If  $A$  is an  $n \times n$  matrix, then a nonzero vector  $\mathbf{x}$  in  $\mathbb{R}^n$  is called an **eigenvector** of  $A$  if  $A\mathbf{x}$  is a scalar multiple of  $\mathbf{x}$ . That is:

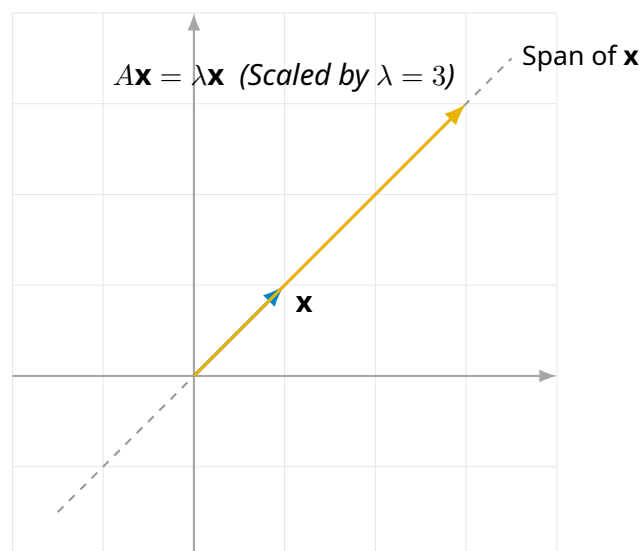
$$A\mathbf{x} = \lambda\mathbf{x}$$

for some scalar  $\lambda$ . The scalar  $\lambda$  is called an **eigenvalue** of  $A$ , and  $\mathbf{x}$  is said to be an eigenvector corresponding to  $\lambda$ .

#### Pro-Tip / Core Intuition

##### Visualizing Eigenvectors

When a linear transformation (a matrix) acts on a vector space, it generally rotates and stretches most vectors off of their original paths. However, an **eigenvector** is a special vector that *does not get knocked off its span* during the transformation. It only gets stretched, squished, or reversed. The factor by which it stretches is the **eigenvalue** ( $\lambda$ ).



### Why This Matters: Real-World Applications

Why are Eigenvectors the most important concept in advanced Computer Science?

- **Google PageRank:** The core algorithm that made Google successful models the internet as a massive matrix. The most important web pages are found by computing the eigenvector associated with the largest eigenvalue of this matrix.
- **Machine Learning (PCA):** Principal Component Analysis (PCA) uses eigenvectors to find the most important features in massive datasets, allowing us to compress data (like images) without losing meaning.
- **Computer Graphics:** 3D engines use eigenvalues to determine axes of rotation and bounding boxes for complex physical simulations.

### The Characteristic Equation

To find the eigenvalues of an  $n \times n$  matrix  $A$ , we rewrite the equation  $A\mathbf{x} = \lambda\mathbf{x}$  as:

$$(\lambda I - A)\mathbf{x} = \mathbf{0}$$

For  $\mathbf{x}$  to be a nonzero vector, the matrix  $(\lambda I - A)$  must be non-invertible. This means its determinant must be zero. Therefore, the eigenvalues of  $A$  are the solutions to the **characteristic equation**:

$$\det(\lambda I - A) = 0$$

When expanded,  $\det(\lambda I - A)$  is a polynomial in  $\lambda$  of degree  $n$ , called the **characteristic polynomial**.

### Eigenspaces and Finding Eigenvectors

Once an eigenvalue  $\lambda$  is found, the corresponding eigenvectors are the non-trivial solutions to the homogeneous linear system:

$$(\lambda I - A)\mathbf{x} = \mathbf{0}$$

The set of all solutions to this system is a subspace of  $\mathbb{R}^n$  called the **eigenspace** corresponding to  $\lambda$ . To find a basis for the eigenspace, substitute the value of  $\lambda$  into the matrix  $(\lambda I - A)$  and use Gaussian elimination (row reduction) to solve for  $\mathbf{x}$ .

### Properties to Remember

- **Triangular Matrices:** If  $A$  is a triangular matrix (upper triangular, lower triangular, or diagonal), then the eigenvalues of  $A$  are simply the entries on its main diagonal.
- **Linear Operators:** If  $T : V \rightarrow V$  is a linear operator, we find its eigenvalues by first finding its standard matrix  $A$  relative to the standard basis, and then finding the eigenvalues of  $A$ .

## Detailed Solutions to Exercise 5.1

### Question 3

**Problem:** Confirm by multiplication that  $\mathbf{x}$  is an eigenvector of  $A$ , and find the corresponding eigenvalue.

$$A = \begin{bmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

**Solution:**

To confirm  $\mathbf{x}$  is an eigenvector, we multiply  $A$  and  $\mathbf{x}$  to see if the result is a scalar multiple of  $\mathbf{x}$  (i.e.,  $A\mathbf{x} = \lambda\mathbf{x}$ ).

$$A\mathbf{x} = \begin{bmatrix} 4 & 0 & 1 \\ 2 & 3 & 2 \\ 1 & 0 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4(1) + 0(2) + 1(1) \\ 2(1) + 3(2) + 2(1) \\ 1(1) + 0(2) + 4(1) \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \\ 5 \end{bmatrix}$$

Factor out the common scalar (5):

$$A\mathbf{x} = 5 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = 5\mathbf{x}$$

Yes,  $\mathbf{x}$  is an eigenvector. The corresponding eigenvalue is  $\lambda = 5$ .

### Question 5(d)

**Problem:** Find the characteristic equation, eigenvalues, and bases for the eigenspaces of the matrix.

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$$

**Solution:**

Characteristic Equation:  $\det(\lambda I - A) = 0$

$$\begin{aligned} \det \begin{bmatrix} \lambda - 1 & 0 \\ -2 & \lambda - 1 \end{bmatrix} &= 0 \\ (\lambda - 1)(\lambda - 1) - (-2)(0) &= 0 \\ (\lambda - 1)^2 &= 0 \end{aligned}$$

**Eigenvalues:**  $\lambda = 1, 1$ .

Now we calculate the eigenbasis by putting  $\lambda = 1$  into  $(\lambda I - A)\mathbf{x} = \mathbf{0}$ :

$$\begin{bmatrix} 1 - 1 & 0 \\ -2 & 1 - 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies \begin{bmatrix} 0 & 0 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Converting the above matrix to a system of equations:

$$-2x_1 = 0 \implies x_1 = 0$$

Let the free variable  $x_2 = t$ .

$$\mathbf{x} = \begin{bmatrix} 0 \\ t \end{bmatrix}$$

Put  $t = 1$ : **Basis:**  $\mathbf{x} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

## Question 7

**Matrix:**

$$A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

**Solution:**

Characteristic Equation:  $\det(\lambda I - A) = 0$

$$\det \begin{bmatrix} \lambda - 4 & 0 & -1 \\ 2 & \lambda - 1 & 0 \\ 2 & 0 & \lambda - 1 \end{bmatrix} = 0$$

Expanding along the second column:

$$(\lambda - 1)[(\lambda - 4)(\lambda - 1) - (-2)] = 0$$

$$(\lambda - 1)[\lambda^2 - 5\lambda + 4 + 2] = 0$$

$$(\lambda - 1)[\lambda^2 - 5\lambda + 6] = 0$$

By middle-term factorization:

$$(\lambda - 1)(\lambda^2 - 3\lambda - 2\lambda + 6) = 0$$

$$(\lambda - 1)(\lambda - 2)(\lambda - 3) = 0$$

**Eigenvalues:**  $\lambda = 1, 2, 3$ .

**For  $\lambda = 1$ :** Substitute into  $(\lambda I - A)$  and reduce to echelon form:

$$\begin{bmatrix} -3 & 0 & -1 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{bmatrix} \begin{array}{l} R_1 \leftrightarrow R_2 \\ R_1 \rightarrow \frac{1}{2}R_1 \end{array} \sim \begin{bmatrix} 1 & 0 & 0 \\ -3 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{array}{l} R_2 \rightarrow R_2 + 3R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Converting to system of equations:

$$x_1 = 0, \quad -x_3 = 0 \implies x_3 = 0$$

Let the free variable  $x_2 = t$ .  $\mathbf{x} = \begin{bmatrix} 0 \\ t \\ 0 \end{bmatrix}$ . Putting  $t = 1$ , **Basis:**  $\mathbf{x} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ .

**For  $\lambda = 2$ :**

$$\begin{bmatrix} -2 & 0 & -1 \\ 2 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 + R_1 \\ R_3 \rightarrow R_3 + R_1 \end{array} \sim \begin{bmatrix} -2 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$R_1 \rightarrow -\frac{1}{2}R_1 \sim \begin{bmatrix} 1 & 0 & 1/2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Converting to system of equations:

$$x_1 + \frac{1}{2}x_3 = 0 \implies x_1 = -\frac{1}{2}x_3$$

$$x_2 - x_3 = 0 \implies x_2 = x_3$$

Let  $x_3 = 2t$  (to clear the fraction). Then  $x_1 = -t$  and  $x_2 = 2t$ .  $\mathbf{x} = \begin{bmatrix} -t \\ 2t \\ 2t \end{bmatrix}$ . Put  $t = 1$ , **Basis:**

$$\mathbf{x} = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}.$$

**For  $\lambda = 3$ :**

$$\begin{bmatrix} -1 & 0 & -1 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 + 2R_1 \\ R_3 \rightarrow R_3 + 2R_1 \end{array} \sim \begin{bmatrix} -1 & 0 & -1 \\ 0 & 2 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{array}{l} R_2 \rightarrow \frac{1}{2}R_2 \\ R_1 \rightarrow -1R_1 \end{array} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Converting to system of equations:

$$x_1 + x_3 = 0 \implies x_1 = -x_3$$

$$x_2 - x_3 = 0 \implies x_2 = x_3$$

Let  $x_3 = t$ . Then  $x_2 = t$ ,  $x_1 = -t$ . Put  $t = 1$ , **Basis:**  $\mathbf{x} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$ .

## Question 9

**Matrix:**

$$A = \begin{bmatrix} 6 & 3 & -8 \\ 0 & -2 & 0 \\ 1 & 0 & -3 \end{bmatrix}$$

**Solution:**

Characteristic Equation:  $\det(\lambda I - A) = 0$

$$\det \begin{bmatrix} \lambda - 6 & -3 & 8 \\ 0 & \lambda + 2 & 0 \\ -1 & 0 & \lambda + 3 \end{bmatrix} = 0$$

Expanding along the second row:

$$\begin{aligned} (\lambda + 2)[(\lambda - 6)(\lambda + 3) - (-1)(8)] &= 0 \\ (\lambda + 2)[\lambda^2 - 3\lambda - 18 + 8] &= 0 \\ (\lambda + 2)[\lambda^2 - 3\lambda - 10] &= 0 \\ (\lambda + 2)[\lambda^2 - 5\lambda + 2\lambda - 10] &= 0 \\ (\lambda + 2)[\lambda(\lambda - 5) + 2(\lambda - 5)] &= 0 \\ (\lambda + 2)(\lambda + 2)(\lambda - 5) &= 0 \end{aligned}$$

**Eigenvalues:**  $\lambda = 5, -2, -2$ .

**For  $\lambda = 5$ :** Reducing to echelon form:

$$\begin{aligned} \begin{bmatrix} -1 & -3 & 8 \\ 0 & 7 & 0 \\ -1 & 0 & 8 \end{bmatrix} & \xrightarrow{R_3 \rightarrow R_3 - R_1} \sim \begin{bmatrix} -1 & -3 & 8 \\ 0 & 7 & 0 \\ 0 & 3 & 0 \end{bmatrix} \\ & \xrightarrow{\substack{R_1 \rightarrow -1R_1 \\ R_2 \rightarrow \frac{1}{7}R_2}} \sim \begin{bmatrix} 1 & 3 & -8 \\ 0 & 1 & 0 \\ 0 & 3 & 0 \end{bmatrix} \\ & \xrightarrow{R_3 \rightarrow R_3 - 3R_2} \sim \begin{bmatrix} 1 & 3 & -8 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Converting to system of equations:

$$x_2 = 0, \quad x_1 + 3x_2 - 8x_3 = 0 \implies x_1 = 8x_3$$

Let  $x_3 = t$ .  $x_1 = 8t$ .  $\mathbf{x} = \begin{bmatrix} 8t \\ 0 \\ t \end{bmatrix}$ . Put  $t = 1$ , **Basis:**  $\mathbf{x} = \begin{bmatrix} 8 \\ 0 \\ 1 \end{bmatrix}$ .

**For  $\lambda = -2$ :** Reducing to echelon form:

$$\begin{aligned} \begin{bmatrix} -8 & -3 & 8 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix} & \xrightarrow{R_1 \leftrightarrow R_3} \sim \begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \\ -8 & -3 & 8 \end{bmatrix} \\ & \xrightarrow{R_3 \rightarrow R_3 - 8R_1} \sim \begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & -3 & 0 \end{bmatrix} \\ & \xrightarrow{R_1 \rightarrow -1R_1} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & -3 & 0 \end{bmatrix} \\ & \xrightarrow{\substack{R_3 \rightarrow -\frac{1}{3}R_3 \\ R_2 \leftrightarrow R_3}} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$



Converting to system of equations:

$$x_2 = 0, \quad x_1 - x_3 = 0 \implies x_1 = x_3$$

Let  $x_3 = t$ .  $x_1 = t$ .  $\mathbf{x} = \begin{bmatrix} t \\ 0 \\ t \end{bmatrix}$ . Put  $t = 1$ , **Basis:**  $\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ .

### Question 10

**Matrix:**

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

**Solution:**

Characteristic Equation:  $\det(\lambda I - A) = 0$

$$\det \begin{bmatrix} \lambda & -1 & -1 \\ -1 & \lambda & -1 \\ -1 & -1 & \lambda \end{bmatrix} = 0$$

Expanding along the first row:

$$\lambda(\lambda^2 - 1) - (-1)(-\lambda - 1) - 1(1 + \lambda) = 0$$

$$\lambda^3 - \lambda - \lambda - 1 - 1 - \lambda = 0$$

$$\lambda^3 - 3\lambda - 2 = 0$$

Put  $\lambda = -1$ :  $-1 + 3 - 2 = 0 \implies 0 = 0$ . By synthetic division for root  $\lambda = -1$ :

$$\begin{array}{r|rrrr} -1 & 1 & 0 & -3 & -2 \\ & & -1 & 1 & 2 \\ \hline & 1 & -1 & -2 & 0 \end{array}$$

The remainder is 0, leaving the quadratic  $\lambda^2 - \lambda - 2 = 0$ .

$$\lambda^2 - 2\lambda + \lambda - 2 = 0$$

$$\lambda(\lambda - 2) + 1(\lambda - 2) = 0$$

$$(\lambda + 1)(\lambda - 2) = 0$$

**Eigenvalues:**  $\lambda = 2, -1, -1$ .

**For  $\lambda = 2$ :** Reducing to echelon form:

$$\begin{aligned} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} R_1 \leftrightarrow R_3 &\sim \begin{bmatrix} -1 & -1 & 2 \\ -1 & 2 & -1 \\ 2 & -1 & -1 \end{bmatrix} \\ R_2 \rightarrow R_2 - R_1 & \\ R_3 \rightarrow R_3 + 2R_1 &\sim \begin{bmatrix} -1 & -1 & 2 \\ 0 & 3 & -3 \\ 0 & -3 & 3 \end{bmatrix} \\ R_3 \rightarrow R_3 + R_2 &\sim \begin{bmatrix} -1 & -1 & 2 \\ 0 & 3 & -3 \\ 0 & 0 & 0 \end{bmatrix} \\ R_1 \rightarrow -1R_1 & \\ R_2 \rightarrow \frac{1}{3}R_2 &\sim \begin{bmatrix} 1 & 1 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Converting to system of equations:

$$x_2 - x_3 = 0 \implies x_2 = x_3$$

$$x_1 + x_2 - 2x_3 = 0 \implies x_1 = 2x_3 - x_2 \implies x_1 = x_3$$

Let  $x_3 = t$ .  $x_2 = t, x_1 = t$ .  $\mathbf{x} = \begin{bmatrix} t \\ t \\ t \end{bmatrix}$ . Put  $t = 1$ , **Basis:**  $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ .

**For  $\lambda = -1$ :** Reducing to echelon form:

$$\begin{aligned} \begin{bmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{bmatrix} R_2 \rightarrow R_2 - R_1 & \\ R_3 \rightarrow R_3 - R_1 &\sim \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ R_1 \rightarrow -1R_1 &\sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Equation:  $x_1 + x_2 + x_3 = 0 \implies x_1 = -x_2 - x_3$ . Let the free variables be  $x_2 = w$  and  $x_3 = t$ .

$\mathbf{x} = \begin{bmatrix} -w - t \\ w \\ t \end{bmatrix}$ . Put  $w = 1, t = 0 \implies \mathbf{x} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ . Put  $w = 0, t = 1 \implies \mathbf{x} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ .

## Question 11

**Matrix:**

$$A = \begin{bmatrix} 4 & 0 & -1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

**Solution:**Characteristic Equation:  $\det(\lambda I - A) = 0$ 

$$\det \begin{bmatrix} \lambda - 4 & 0 & 1 \\ 0 & \lambda - 3 & 0 \\ -1 & 0 & \lambda - 2 \end{bmatrix} = 0$$

Expanding along the second row:

$$\begin{aligned} (\lambda - 3) [(\lambda - 4)(\lambda - 2) - (-1)(1)] &= 0 \\ (\lambda - 3) [\lambda^2 - 6\lambda + 8 + 1] &= 0 \\ (\lambda - 3) [\lambda^2 - 3\lambda - 3\lambda + 9] &= 0 \\ (\lambda - 3) [\lambda(\lambda - 3) - 3(\lambda - 3)] &= 0 \\ (\lambda - 3)(\lambda - 3)(\lambda - 3) &= 0 \end{aligned}$$

**Eigenvalues:**  $\lambda = 3, 3, 3$ .**For**  $\lambda = 3$ : Reducing to echelon form:

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \sim \begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_1 \rightarrow -1R_1} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Converting to system of equations:

$$x_1 - x_3 = 0 \implies x_1 = x_3$$

Let  $x_3 = t$  and  $x_2 = w$ .  $x_1 = t$ .  $\mathbf{x} = \begin{bmatrix} t \\ w \\ t \end{bmatrix}$ . Put  $t = 1, w = 0 \implies \mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ . Put  $t = 0, w = 1 \implies$ 

$$\mathbf{x} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}.$$

**Question 12****Matrix:**

$$A = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$$

**Solution:**Characteristic Equation:  $\det(\lambda I - A) = 0$ 

$$\det \begin{bmatrix} \lambda - 1 & 3 & -3 \\ -3 & \lambda + 5 & -3 \\ -6 & 6 & \lambda - 4 \end{bmatrix} = 0$$

Expanding along the first row:

$$\begin{aligned}
 &\Rightarrow (\lambda - 1)[(\lambda + 5)(\lambda - 4) - 18] - 3[-3(\lambda - 4) - 18] - 3[-18 - (-6)(\lambda + 5)] = 0 \\
 &\Rightarrow (\lambda - 1)[\lambda^2 + \lambda - 2] - 3[-3\lambda - 6] - 3[6\lambda + 12] = 0 \\
 &\Rightarrow (\lambda - 1)(\lambda + 2)(\lambda - 1) + 9(\lambda + 2) - 18(\lambda + 2) = 0 \\
 &\Rightarrow (\lambda + 2)[(\lambda - 1)^2 + 9 - 18] = 0 \\
 &\Rightarrow (\lambda + 2)[\lambda^2 - 2\lambda - 8] = 0 \\
 &\Rightarrow (\lambda + 2)[\lambda^2 - 4\lambda + 2\lambda - 8] = 0 \\
 &\Rightarrow (\lambda + 2)[\lambda(\lambda - 4) + 2(\lambda - 4)] = 0 \\
 &\Rightarrow (\lambda + 2)(\lambda + 2)(\lambda - 4) = 0
 \end{aligned}$$

**Eigenvalues:**  $\lambda = 4, -2, -2$ .

**For  $\lambda = 4$ :** Reducing to echelon form:

$$\begin{aligned}
 \begin{bmatrix} 3 & 3 & -3 \\ -3 & 9 & -3 \\ -6 & 6 & 0 \end{bmatrix} & \begin{matrix} R_1 \rightarrow \frac{1}{3}R_1 \\ R_2 \rightarrow \frac{1}{3}R_2 \\ R_3 \rightarrow \frac{1}{3}R_3 \end{matrix} \sim \begin{bmatrix} 1 & 1 & -1 \\ -1 & 3 & -1 \\ -2 & 2 & 0 \end{bmatrix} \\
 \begin{matrix} R_2 \rightarrow R_2 + R_1 \\ R_3 \rightarrow R_3 + 2R_1 \end{matrix} & \sim \begin{bmatrix} 1 & 1 & -1 \\ 0 & 4 & -2 \\ 0 & 4 & -2 \end{bmatrix} \\
 \begin{matrix} R_2 \rightarrow \frac{1}{4}R_2 \\ R_3 \rightarrow R_3 - R_2 \end{matrix} & \sim \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{bmatrix} \\
 R_1 \rightarrow R_1 - R_2 & \sim \begin{bmatrix} 1 & 0 & -1/2 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{bmatrix}
 \end{aligned}$$

Converting to system of equations:

$$x_2 - \frac{1}{2}x_3 = 0 \implies x_2 = \frac{1}{2}x_3$$

$$x_1 - \frac{1}{2}x_3 = 0 \implies x_1 = \frac{1}{2}x_3$$

Let  $x_3 = 2t$ .  $x_1 = t, x_2 = t$ .  $\mathbf{x} = \begin{bmatrix} t \\ t \\ 2t \end{bmatrix}$ . Put  $t = 1$ , **Basis:**  $\mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ .

**For  $\lambda = -2$ :** Reducing to echelon form:

$$\begin{aligned}
 \begin{bmatrix} -3 & 3 & -3 \\ -3 & 3 & -3 \\ -6 & 6 & -6 \end{bmatrix} & \begin{matrix} R_1 \rightarrow -\frac{1}{3}R_1 \\ R_2 \rightarrow -\frac{1}{3}R_2 \\ R_3 \rightarrow -\frac{1}{6}R_3 \end{matrix} \sim \begin{bmatrix} 1 & -1 & 1 \\ 1 & -1 & 1 \\ 1 & -1 & 1 \end{bmatrix} \\
 \begin{matrix} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{matrix} & \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}
 \end{aligned}$$

Converting to system of equations:

$$x_1 - x_2 + x_3 = 0 \implies x_1 = x_2 - x_3$$

Let  $x_2 = w$  and  $x_3 = t$ .  $x_1 = w - t$ .  $\mathbf{x} = \begin{bmatrix} w - t \\ w \\ t \end{bmatrix}$ . Put  $w = 1, t = 0 \implies \mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ . Put  $w = 0, t = 1 \implies \mathbf{x} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ .

### Question 13

**Problem:** Find the characteristic equation by inspection.

$$A = \begin{bmatrix} 3 & 0 & 0 \\ -2 & 7 & 0 \\ 4 & 8 & 1 \end{bmatrix}$$

**Solution:**

Because  $A$  is a lower triangular matrix, its eigenvalues are simply the entries on its main diagonal. The eigenvalues are  $\lambda = 3, 7, 1$ . Therefore, the characteristic equation is:

$$(\lambda - 3)(\lambda - 7)(\lambda - 1) = 0$$

### Question 16

**Problem:** Find the eigenvalues and a basis for each eigenspace of the linear operator.

$$T(x, y, z) = (2x - y - z, x - z, -x + y + 2z)$$

**Solution:**

Extracting the standard matrix  $A$ :

$$A = \begin{bmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ -1 & 1 & 2 \end{bmatrix}$$

Characteristic Equation:  $\det(\lambda I - A) = 0$

$$\det \begin{bmatrix} \lambda - 2 & 1 & 1 \\ -1 & \lambda & 1 \\ 1 & -1 & \lambda - 2 \end{bmatrix} = 0$$

$$\begin{aligned}
 \Rightarrow (\lambda - 2)[\lambda(\lambda - 2) - (-1)(1)] - 1[(-1)(\lambda - 2) - (1)(1)] + 1[(-1)(-1) - \lambda(1)] &= 0 \\
 \Rightarrow (\lambda - 2)[\lambda^2 - 2\lambda + 1] - 1[-\lambda + 2 - 1] + 1[1 - \lambda] &= 0 \\
 \Rightarrow (\lambda - 2)(\lambda - 1)^2 - (-\lambda + 1) + (1 - \lambda) &= 0 \\
 \Rightarrow (\lambda - 2)(\lambda - 1)^2 + (\lambda - 1) - (\lambda - 1) &= 0 \\
 \Rightarrow (\lambda - 2)(\lambda - 1)^2 &= 0
 \end{aligned}$$

**Eigenvalues:**  $\lambda = 1, 1, 2$ .

**For  $\lambda = 1$ :** Reducing to row echelon form:

$$\begin{aligned}
 \begin{bmatrix} -1 & 1 & 1 \\ -1 & 1 & 1 \\ 1 & -1 & -1 \end{bmatrix} & \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 + R_1 \end{array} \sim \begin{bmatrix} -1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\
 & R_1 \rightarrow -1R_1 \sim \begin{bmatrix} 1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}
 \end{aligned}$$

Converting to system of equations:

$$x_1 - x_2 - x_3 = 0 \implies x_1 = x_2 + x_3$$

Let the free variables be  $x_2 = w$  and  $x_3 = t$ .  $x_1 = w + t$ .  $\mathbf{x} = \begin{bmatrix} w+t \\ w \\ t \end{bmatrix}$ . Put  $w = 1, t = 0 \implies$

$$\mathbf{x} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}. \text{ Put } w = 0, t = 1 \implies \mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

**For  $\lambda = 2$ :** Reducing to strictly defined echelon form:

$$\begin{aligned}
 \begin{bmatrix} 0 & 1 & 1 \\ -1 & 2 & 1 \\ 1 & -1 & 0 \end{bmatrix} & R_1 \leftrightarrow R_3 \sim \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix} \\
 & R_2 \rightarrow R_2 + R_1 \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \\
 & R_3 \rightarrow R_3 - R_2 \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \\
 & R_1 \rightarrow R_1 + R_2 \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}
 \end{aligned}$$

Converting to system of equations:

$$x_2 + x_3 = 0 \implies x_2 = -x_3$$

$$x_1 + x_3 = 0 \implies x_1 = -x_3$$

Let the free variable  $x_3 = t$ .  $x_2 = -t, x_1 = -t$ .  $\mathbf{x} = \begin{bmatrix} -t \\ -t \\ t \end{bmatrix}$ . Put  $t = 1 \implies$  **Basis:**  $\mathbf{x} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$ .

## About This Compilation

---

### The Motive

This document is not a standalone textbook, but rather a carefully curated compilation of all the core topics, lecture questions, and assignments covered during my third-semester Linear Algebra course, taught by **Dr. Fozia Hanif** at UBIT.

University-level mathematics can be rigorous and demanding. By consolidating the foundational theory, theorems, and step-by-step problem-solving techniques into a single, comprehensive resource, my primary goal is to pass the torch to future students navigating this exact course. If you are taking this class, this guide is designed to help you save valuable time, avoid the struggle of hunting down scattered materials, and prepare for your exams with absolute confidence.

### Join The UBIT Hub Community

**Let's grow together!** The UBIT Hub is a dedicated WhatsApp community built for our students. Its primary aim is to share high-quality notes, learning resources, and help everyone succeed together without having to struggle for the best material. Whether you are solving complex matrices or preparing for midterms, you don't have to do it alone.

[Click Here to Join The UBIT Hub on WhatsApp](#)

### Let's Connect & Collaborate

**LINKEDIN** // [codewithmahad](#)

**GITHUB** // [codewithmahad](#)

*Built for the community. Share freely.*